

College Algebra (Notes)

natalie <3

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– INTRODUCTORY TEXT –

(Please read for more information!)

Everything preceding this text considers the reader to know basic logic (propositions, truth values, conjunctions etc.) as that kind of notation will be seen. Another prerequisite is for basic set theory, as we will be mentioning sets often. There are great reads online. All of these are observed by the varying dates. Just note, not all topics will be covered. The contents of these notes have the topic, date, and discussion involved, respectively. Since the dates is the date when I started noting down, the full date is listed at the end of each sub-sub-section within "Finishing Remarks." *Date format is dd/mm/yy, and time is in military time.* We will assume that the reader knows the prerequisites learnt prior. Note that, these notes stretch across various books, meaning they are not from one author. A list of books will be provided at the end. Some user-generated content (like examples, definitions, or sub-content) may be given. The table of contents are listed precedingly.

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1 Real Numbers

As a pre-introduction, or my thoughts on this topic: All real numbers are important. Either rational or irrational, it is the main form of how every amount works. However, this will be a more formal introduction to the reals. Rather than introducing the discussion with just pre-made descriptions, we will provide the basic axioms of each of the real numbers as it will be a more formal description of the reals. Therefore, theorems, and axioms will be shown. All proofs will be left as a thought or discussion, or exercise for the reader.

1.1 April 10, 2024.

The term of a *number* is an abstract definition. There are two systems which describe symbols for all numbers, the *Hindu-Arabic System*, or the *Roman Numeral System*. The Roman Numeral System is unused for most of modern mathematics, as it is considered a legacy definition. Nowadays, the Hindu-Arabic system is primary.

The Hindu-Arabic system, is described with ten symbols: 1, 2, 3, 4, 5, 6, 7, 8, 9, 0. The Roman Numeral System, is described with seven symbols: I, V, X, L, C, D, M. Definitions for the Roman Numeral System will not be provided as we will use the system of Hindu-Arabic.

A *number system* is a set of numbers together with the binary operations. A *binary operation* is the process of taking any two numbers at a time, and doing operations on them. Note that a number is a concept, and a numeral is a symbol for a number.

1.1.1 Concepts of a Mathematical System

The system of real numbers is considered a *mathematical system* or *structure*. In any system, there are rules for it to work. For the system of the reals, they are axioms, propositions, notions, corollaries, lemmas, theorems, and definitions. As any system, they have *cyclic definitions*. These kinds of definitions are undefined and recursive. If item A has a definition that relies on B , and B is a definition that relies on A , it is a cyclic definition. Definitions are either shall not be proven or disproven, neither defended.

An *axiom* or a *postulate* is a definition that is accepted by assumption or without proof. Therefore, they need not be proven or with defence. A *proof* is an argument or reason that is based off of axioms, or theorems. A *theorem* is a statement with a truth value that is with proof. Like the Pythagorean Theorem, it has a basis of reason through postulates. In geometry, axioms are postulates. Before we conclude, a set of *undefined terms* or *primitive terms* must be needed. They can be cyclic, or unknown. Like a number, they can not be defined. Leaving some terms undefined and accepting them is needed for a system. Step by step, in a system:

1. A set of undefined terms
2. A set of defined terms
3. A set of axioms
4. A set of theorems

1.1.2 History of the Real Numbers

All subsets of the real numbers are described. For now, we will assume these are the defined subsets of the reals. A diagram can be represented but will be needless for now. All subsets possible in the real numbers that are described:

- $\mathbb{N}$ = *Natural Numbers* - all possible counting numbers.
- $\mathbb{Z}$ = *Integers* - all numbers that are whole, including the negatives below zero.

- $\mathbb{Q} = \text{Rational Numbers}$ - all numbers that are represented as a ratio, or a fraction.
- $\mathbb{P} = \text{Irrational Numbers}$ - all numbers that are incapable of representing as a ratio or a fraction.
- $\mathbb{R} = \text{Real Numbers}$ - the entire set of real numbers.

1.1.3 System of Natural numbers

These numbers are represented as this set:

$$\mathbb{N} = \{1, 2, 3, 4, \dots\}$$

There are five axioms that are described with the system of natural numbers.

1. AXIOM 1: *CLOSURE*:
 - a) If $a, b \in \mathbb{N}$, $\Rightarrow a + b \in \mathbb{N}$.
 - b) If $a, b \in \mathbb{N}$, $\Rightarrow ab \in \mathbb{N}$.
2. AXIOM 2: *COMMUTATIVITY*:
 - a) If $a, b \in \mathbb{N}$, $\Rightarrow a + b = b + a$.
 - b) If $a, b \in \mathbb{N}$, $\Rightarrow ab = ba$.
3. AXIOM 3: *ASSOCIATIVITY*:
 - a) If $a, b, c \in \mathbb{N}$, $\Rightarrow (a + b) + c = a + (b + c)$.
 - b) If $a, b, c \in \mathbb{N}$, $\Rightarrow (ab)c = a(bc)$.
4. AXIOM 4: *DISTRIBUTIVITY*:
 - a) If $a, b, c \in \mathbb{N}$, $\Rightarrow a(b + c) = ab + ac$.
5. AXIOM 5: *IDENTITY*:
 - a) If $\exists a \in \mathbb{N}$, $\Rightarrow a \times 1 = a = 1 \times a = a$.

Axiom 5a is described as the *multiplicative identity* of a number a .

1.1.4 System of Integers

If $a, b \in \mathbb{N}$ therefore $a - b$ is limited to if a is smaller than b . However, if no limitation would be there, the system would be classified as the *set of integers*. Described as:

$$\mathbb{Z} = \{\dots - 2, -1, 0, 1, 2, \dots\}$$

Integers are represented with $\mathbb{Z}$. Some textbooks refer to it with an $\mathbb{I}$, like the one I am noting, however, we will assume it is. Integers have the number zero, and the negatives. These are all the elements not found in the natural numbers.

With this knowledge, there will be two new axioms, introduced with the system of integers.

1. AXIOM 5: *IDENTITY*:
 - b) If $\exists a \in \mathbb{Z}$, $\Rightarrow a + 0 = a = 0 + a$.
2. AXIOM 6: *INVERSE*:
 - a) If $\forall a \in \mathbb{Z}$, $\Rightarrow a + (-a) = 0 = -a + a$.

As a continuation, Axiom 5b is called the *additive identity*. In Axiom 6a, it is called the *additive inverse*. In the system of integers, we can define the definition of subtraction.

Definition 1.1 (Subtraction). If $a, b \in \mathbb{Z}$, $\Rightarrow a - b = a + (-b)$.

1.1.5 System of Rational Numbers

As a definition.

Definition 1.2 (Division). If $a, b, c \in \mathbb{Z} \wedge b \neq 0, \Rightarrow \frac{a}{b} \vee a \div b = c, \Leftrightarrow a = bc$

A definition for division can derive the next definition, a rational number.

Definition 1.3 (Rational Numbers). If $a, b \in \mathbb{Z} \wedge b \neq 0, \Rightarrow \frac{a}{b}$ is a rational number.

If we considered $b = 0$, then division by zero is impossible. As an example: *If you have 15 items, and you want to share it with zero people, how will you share it? If you share it still, where are the people?* We will continue to obey this law by definition, and if we were to have a ratio wherein $b = 0$, then the description is *undefined*.

A new axiom can also be derived, as a continuation for Axiom 6, inverses.

1. AXIOM 6: *INVERSE*:

b) $\forall a \in \mathbb{Q} \wedge a \neq 0, \exists b = a$ wherein $(a)(\frac{1}{b}) = 1 = (\frac{1}{b})(a)$.

We can derive a new definition, that of division. Note that division *is closed* under the set of rational numbers. As for division, it is defined through multiplication.

Definition 1.4 (Defining Division). If $a, b \in \mathbb{Q} \wedge b \neq 0, \Rightarrow \frac{a}{b} = a \times \frac{1}{b}$.

FINISHING REMARKS – *The next factor for this topic will be in the next part, which is considered separate due to its nature, considered irrationals and reals. As this chapter departs, all topics within this time frame are considered done. Tomorrow will be when the note-taking will resume, and when all will be more classified.*

Date of Beginning - 10/04/24, 23:17.

Date of End - 12/04/24, 12:06.

1.2 April 12, 2024.

The following text will include the topic irrational and real numbers. It will also include theorems, as well partaking in proving their truth, with using the axioms of the reals. This text may take a bit longer to understand, but it may be doable with a bit of research on proof-making. As much as this topic involves proof, we must take note that a prerequisite of inductive and deductive reasoning must be made, or have some knowledge of proof-making, to continue.

1.2.1 System of Irrational Numbers

An *irrational number* is a number which can't be expressed as a common ratio. The most common type of irrational number is a square root. $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$ and etc, are all considered irrational roots. They typically have an infinite amount of decimals. Certain square roots are omitted because they have a rational value. For example, $\sqrt{4} = 2$. Note that, the symbol for irrationals are not really properly defined in modern textbooks mostly, but we will assume it is $\mathbb{P}$. As a set:

$$\mathbb{P} = \{x | x \in \mathbb{R} \wedge x \notin \mathbb{Q}\}$$

1.2.2 System of Real Numbers

Defining the set of real numbers ($\mathbb{R}$) is through:

Definition 1.5 (The Set of the Real Numbers). $\mathbb{R} = \mathbb{Q} \cup \mathbb{P}$

All axioms apply to the real numbers, and we can rewrite these axioms with respect to the real numbers, rather than in respect with the specific subsets they apply in locally. To save time, rewriting won't be necessary. However, there will be six newer axioms in the set of reals, that will encompass all sets. Called *equality axioms*, described:

1. EQUALITY AXIOM 1: *REFLEXIVE*:

$$\forall a \in \mathbb{R}, \Rightarrow a = a.$$

2. EQUALITY AXIOM 2: *SYMMETRY*:

$$\forall a, b, \Rightarrow a = b \Leftrightarrow b = a.$$

3. EQUALITY AXIOM 3: *TRANSITIVITY*:

$$\forall a, b, c \in \mathbb{R}, \text{ wherein } a = b \wedge b = c, \Rightarrow a = c.$$

4. EQUALITY AXIOM 4: *SUBSTITUTION*:

$\forall a, b \in \mathbb{R} \wedge a = b$, a or b could be substituted to any equation that has a and b and will not be changed.

5. EQUALITY AXIOM 5: *ADDITION PROPERTY*:

$$\forall a, b, c \in \mathbb{R} \wedge a = b, \Rightarrow a + c = b + c.$$

6. EQUALITY AXIOM 6: *MULTIPLICATION PROPERTY*:

$$\forall a, b, c \in \mathbb{R} \wedge a = b, \Rightarrow ac = bc.$$

In the field of the reals, we can describe our first 18 theorems from our twelve postulates/axioms (including the first six, and the equality axioms.) We can describe the following theorems:

Theorem 1.1 (Extension of Addition Property). $a = b \wedge c = d, \Rightarrow a + c = b + d$

Theorem 1.2 (Extension of Multiplication Property). $a = b \wedge c = d, \Rightarrow ac = bd$

Theorem 1.3 (Converse of Addition Property). $a + c = b + c \Rightarrow a = b$

Theorem 1.4 (Converse? of Multiplication Property). $ac = bc \Rightarrow a = b \Leftrightarrow c \neq 0$

Theorem 1.5 (Multiplication of Zero). $a \times 0 = 0 = 0 \times a$

Theorem 1.6 (Defining Multiplying by Zero). $ab = 0 \Rightarrow a = 0 \vee b = 0$

Theorem 1.7 (Negative Product Laws # 1). $-(-a) = a$

Theorem 1.8 (Negative Product Laws # 2). $(-1)(a) = -a$

Theorem 1.9 (Negative Product Laws # 3 & # 4). $(-a)(b) \vee (a)(-b) = -(ab)$

Theorem 1.10 (Negative Product Laws # 5). $(-a)(-b) = ab$

Theorem 1.11 (Negative Quotient Laws # 1 & # 2). $\frac{a}{-b} \vee \frac{-a}{b} = -\frac{a}{b} \Leftrightarrow b \neq 0$

Theorem 1.12 (Negative Quotient Laws # 3). $\frac{-a}{b} = -\frac{a}{b} \Leftrightarrow b \neq 0$

Theorem 1.13 (Distributing a Negative # 1). $-(a + b) = -a - b$

Theorem 1.14 (Distributing a Negative # 2). $-(a - b) = -a + b$

Theorem 1.15 (Extension of Distributivity). $a(b - c) = ab - ac$

Theorem 1.16 (Dividing by Itself). $\frac{a}{a} = 1 \Leftrightarrow a \neq 0$

Theorem 1.17 (Defining Zero Divided by Anything). $\frac{0}{a} = 0 \Leftrightarrow a \neq 0$

Theorem 1.18 (Dividing by One). $\frac{a}{1} = a$

Theorem 1.18 is also how we can define the set of integers as rational numbers. We can prove some of the following, through the various axioms we have introduced beforehand. We will prove Theorem 1.3, 1.7, 1.9, 1.10, and 1.16 as examples. Note that these are proofs of these theorems, not examples of what these theorems are. To prove, one must know how to deduct or induct reason from axioms given.

We will start off with the proofs:

1) **THEOREM 1.3.**

Proof. Given: $a+c = b+c \Rightarrow a = b$. Using the *Addition Property of Equality*, $(a+c)+(-c) = (b+c)+(-c)$. Through *Definition 1.1*, $(a+c)-c = (b+c)-c$. Through *Associativity*, $a+(c-c) = b+(c-c)$. Through *Inverse*, $a+0 = b+0$. Then, as *Identity*, $a+0 = a$, and $b+0 = b$. As a result, $a = b$. $\square$

This is the first example of us using a square as our *quod erat demonstrandum*, or QED, to end our proof. We will assume this.

2) **THEOREM 1.7.**

Proof. Given: $-(-a) = a$. Note that, if we use the *Inverse* of addition, $a + (-a) = 0$. Through *Substitution* from $-(-a) = a$, however with *Symmetry*, $a = -(-a)$, therefore substitute as a . Therefore, with substitution, $-a + (-(-a)) = 0$. Noting that, since $a = -(-a)$, we can infer that $-a + (-(-a)) = 0 = -a + a = 0$. Using *Commutativity*, $a + (-a) = 0$. Since we used that definition from our proof, as a result, $-(-a) = a$. $\square$

This is not a rigorous proof, however it is mostly detailed with definitions but can be accepted through different means.

3) **THEOREM 1.9.**

Proof. Given: $(-a)(b) = -ab$. Note that, with *Symmetry*, $-ab = (-a)(b)$. Using *Substitution*, $(-a)(b) = (-a)(b)$ and holds true due to the *Reflexive Property*. We will note that, on one side, say the right, we know through *Identity*, $(-a)(b) = (-a)(b) + 0$. Using this information, 0 can be defined through the *Identity*, therefore $(-a)(b) = (-a)(b) + (((-a)(b)) + (-(-a) + (-b)))$. Since we know that $-ab = (-a)(b)$, reverse it using *Reflexive* so $(-a)(b) = -ab$. With *Substitution*, $(-a)(b) = (-a)(b) + ((ab) + (-ab))$. Due to *Associativity*, $(-a)(b) = ((-a)(b) + ab) + (-ab)$. Thereby using *Distributivity*, $(-a)(b) = b((-a) + a) + (-ab)$. With this definition, note with *Identity* again, that $(-a) + a = 0$. Inferring this, $(-a)(b) = b(0) + (-ab)$. With *Theorem 1.5*, note that $b \times 0 = 0$. With this, $(-a)(b) = 0 + (-ab)$. Finally, with the *Identity*, $0 + (-ab) = -ab$. As a result, $(-a)(b) = -ab$. $\square$

Again, not a really rigorous proof. However, notice that we use *Theorem 1.5* as one of our proofs. However, we must note that, since Theorem 1.9 is included here, we must assume that the proof has already been described. From now on, we will derive the following with respect to using theorems included.

4) **THEOREM 1.10**

Proof. Given: $(-a)(-b) = ab$. Note that, using *Theorem 1.8*, we can tell that $(-1 \times a)(-1 \times b) = ab$. Rather than using distributivity, we can project to use *Associativity* and *Commutativity*, since we can re-arrange -1 and move them to a separate bracket. Therefore, $(-1)(-1)(ab)$, simplified $a \times b$ to ab . Using *Theorem 1.7*, note that $(-1)(-1) = 1$. Hence, $1(ab) = ab$. Through *Identity*, $1(ab) = ab$. Therefore, we get $ab = ab$. Hence, this holds true. $\square$

This proof is often not really defined. We typically associate this proof with *Theorem 1.7* as it is closely related and tied to.

5) **THEOREM 1.16**

Proof. Given: $\frac{a}{a} = 1 \Rightarrow a \neq 0$. With *Definition 1.2*, $\frac{a}{a} = (a)(\frac{1}{a})$. Using *Inverse*, $(a)(\frac{1}{a}) = 1$. Therefore, as a result, $1=1$. Hence, it holds true. $\square$

This is the smallest proof across previous.

FINISHING REMARKS – We have concluded a major part of the system of real numbers. We have proved five theorems with inductive reasoning. The following will include decimal conversions,

factors, multiples, and prime numbers, and indices, a.k.a exponents. This might take three days to cover, as an estimate. Rather, there will be a gap to answer exercises that were given. With this departure, I will appreciate all the inductive and deductive reasoning provided. It really took me forever to think, and most may be wrong (but may be correct.) On the 17th is when this will resume.

Date of Beginning - 12/04/24, 9:44.

Date of End - 14/04/24, 21:51.

1.3 April 17, 2024.

[21:01] Now including start time, rather than estimation. The following will endure on what I have said for this section. It will include decimal conversions, factors, multiples, primes numbers, and indices or exponents. Note that, this section is a more formal introduction. Expect some introductory sequences sound vague. As forth, we shall move forward.

1.3.1 A Light Intro to Decimals

A *decimal* is a numerical extension of a number, denoted by a dot (.). A decimal is a small increment to a whole number. As a connotation, decimals do exist in whole numbers. Take the following:

$$19 = 19.0000...$$

$$\frac{1}{2} = 0.50000...$$

These are various definitions on decimals. Note that, it is a most common notion to omit the decimal repetition to save space. Consider $\frac{1}{2} = 0.5000....$ We can shorten it to 0.5 rather than extending a list of zeros. Consider 12.16. Such decimal that has an ending is called a *terminating* decimal. It does not go on endlessly. A decimal that does on with no end is called a *non-terminating* decimal.

A non-terminating decimal can have two definitions. The term *periodic* means it is a pattern. The definitions are:

- *Non-Terminating and Periodic* - It is a rational expression.
- *Non-Terminating and Non-Periodic* - It is irrational.

An example of a non-terminating and non-periodic decimals can be: $\sqrt{2} = 1.414213..., e = 2.18281828459$, or $\pi = 3.14159265359...$ and more.

1.3.2 Terminating Decimal Conversion

Decimals have various parts. Take note in, say, 9.4192. The 9 is the *whole number part*. The 4 is the *tenths* part, the 1 is the *hundredths* part, etc.

To convert a terminating decimal to a fraction, consider its structure. If it ends at the tenths place, the denominator will be a 10. If it ends at a hundredths, the denominator will be a 100, and so forth. Examples given:

- Convert 0.5.
 1. Take note of the structure. The last of the decimal is the tenths place. The denominator will be a 10.
 2. Put everything on the decimal part as the numerator. Therefore, $\frac{5}{10}$.
 3. Reduce to lowest terms if possible. Since 5 and 10 are divisible by 5, convert to lowest terms by dividing the top and bottom by 5 separately. Therefore, $\frac{5 \div 5}{10 \div 5}$. This will become $\frac{1}{2}$.

Therefore, it is $\frac{1}{2}$.

- Convert 0.33.

1. Take note of the structure. The last of the decimal is the hundredths place. The denominator will be a 100.
2. Put everything on the decimal part as the numerator. Therefore, $\frac{33}{100}$.
3. Reduce to lowest terms if possible. Since the fraction can't be divided by anything related to a number that can divide both top and bottom, and is the biggest number that can do so, the fraction will remain as $\frac{33}{100}$.

Therefore, it is $\frac{33}{100}$.

- Convert 2.35.

1. Take note of the structure. Our previous considerations have 0 as the whole number part. We will ignore the whole number part for now. The last decimal is the hundredths place. The denominator will be a 100.
2. Put everything on the decimal part as the numerator. Therefore, $\frac{35}{100}$.
3. Reduce to lowest terms if possible. 35 and 100 are both divisible by 5. Therefore, it will become $\frac{7}{20}$.
4. Consider the whole number part. Firstly, take the reduced denominator and multiply to the whole number part. Therefore, $2 \times 20 = 40$.
5. Add the reduced fraction and the whole number part that was multiplied by the denominator. So, $40 + 7 = 47$.
6. Set this sum as the new numerator with retaining the denominator. So, it will result to $\frac{47}{20}$.
7. Reduce if possible. Since it is not reducible, it will retain as is.

Therefore, it is $\frac{47}{20}$.

This is a definite conversion for decimals which have an ending. Decimal structure is of course, objective. It can have a ten-thousandths, or a millionths place. Despite this, by doing the order given it is in simple manner that can be provided to convert such to fractions. However, recurring (non-terminating) decimal places that have no end are always subject to different forms. They can be periodic and non-periodic. However, we will let this up to the next section to discussion.

1.3.3 Non-Terminating Decimal Conversion

WITH NO NON-REPEATING DIGIT IN DECIMAL: Instead of using 10, or 100, or 1000, we use 9, 99, or 999, etc. The amount of 9's is the amount of digits in the repeating part.

In the case of notation wise, to save horizontal space, we must introduce the *vinculum*. It is a horizontal bar that can be put under or over. It once served for parentheses, but modernized to be mostly seen in decimals. If a and b are the whole and decimal part respectfully, a non-terminating decimal would be $a.\bar{b}$. To convert such, with examples:

- Convert $0.\bar{5}$.

1. Take note of the structure, and count how many digits the repeated part has. It has one digit. Therefore, the denominator will be a 9.
2. Put everything on the decimal part as the numerator. Therefore, $\frac{5}{9}$.
3. Reduce to lowest terms if possible. Since 5 and 9 are not reducible by a common number and is not the biggest of each, it will remain as so.

Therefore, it is $\frac{5}{9}$.

- Convert $4.\overline{48}$.
 1. Take note of the structure, and count how many digits the repeated part has. Ignore the whole number part for now. It has two digits. Therefore, the denominator will be a 99.
 2. Put everything on the decimal part as the numerator. Therefore, $\frac{48}{99}$.
 3. Reduce to lowest terms if possible. Since 48 and 99 are divisible by 3, the fraction will become $\frac{16}{33}$.
 4. Consider the whole number part. Take the reduced denominator and multiply it by the whole number part, $4 \times 33 = 132$.
 5. Add the reduced fraction and the whole number part that was multiplied by the denominator. Thereby $132 + 16 = 148$.
 6. Set this sum as the new numerator with retaining the denominator. So, it will be $\frac{148}{33}$.
 7. Reduce if possible. Since it is not reducible, it will retain as is.

Therefore, it is $\frac{148}{33}$.

WITH A NON-REPEATING DIGIT IN DECIMAL: If we were to consider a, b and c as the whole number part (a) and decimal part (b and c), then if the decimal expression is expressed as $a.b\overline{c}$, the conversion may be explained by one example in the following:

- Convert $1.34\overline{72}$.
 1. Count how many non-repeating numbers there are. Ignore the whole number part for now. In $1.34\overline{72}$, there are two numbers non-repeating.
 2. Count how many repeating numbers there are in the decimal part. In $1.34\overline{72}$, there are two numbers that are repeating.
 3. Create a number with 9's and 0's for our denominator. The amount of non-repeating numbers are the amount of zeros, and the amount of repeating numbers are the amount of nines. Therefore, since there are two repeating and non-repeating, it becomes 9900.
 4. Take the decimal part and subtract by the non-repeating digits, to make the numerator. Ignore the overline, and consider it as a normal number. $1.34\overline{72}$, $34\overline{72}$ is just 3472. Then, $3472 - 34 = 3438$.
 5. Create the fraction. It becomes $\frac{3438}{9900}$.
 6. Reduce if possible. 3438 and 9900 are both divisible by 18. Therefore, it becomes $\frac{191}{550}$.
 7. Take the whole number part, and multiply by the denominator (reduced in this case.) Therefore, $1 \times 550 = 550$.
 8. Add this number to the numerator (reduced in this case.) Thereby $550 + 191 = 741$.
 9. Set that as the new numerator, and reduce if possible. Therefore, it will become $\frac{741}{550}$.

Therefore, it is $\frac{741}{550}$.

Irrational expressions are numbers that are non-terminating, and non-periodic. They do not have a pattern. These are the most famous mathematical constants, or roots that do not have a perfect exponential.

1.3.4 Factors, Multiples, and Primes

A more formal definition of a multiple and a factor is:

Definition 1.6. If $a, b \in \mathbb{N}$, $\exists x \in \mathbb{N}$ such that $ax = b$, then a is a factor of b and b is said to be a multiple of a .

To put it simply, a factor is a number that can divide b . A multiple is a number that is multiplied by a number x to a . Every natural number has 1 as a factor. The set of all factors for 12 is $\{1, 2, 3, 4, 6, 12\}$. All factors that are *not* 1 are called the *proper factors*.

The definition for even and odd numbers are:

Definition 1.7. $\exists n \in \mathbb{Z} \wedge n$ is divisible by 2, then n is even. As another definition, $\exists n \in \mathbb{Z} \wedge n$ is not divisible by 2, then n is odd.

Mathematically, if represented by formulas, the definition would be:

Definition 1.8. Let $k \in \mathbb{Z}$, and n is a multiple of k . If $n = 2k$, n is even. If $n = 2k + 1$, n is odd. If dividing an odd number n by 2, there will be a remainder of 1.

However, there exists such numbers which do not exist as multiples of other numbers whatsoever. A definition:

Definition 1.9. $\exists p \in \mathbb{Z}$. p is prime if and only if $p > 1 \wedge p$ has only two factors, itself and 1.

As an observation, every prime number is *odd* except for 2. 2 has only two factors, 2 and 1. Note that, every prime is odd, but every odd number is not prime. 13 is prime, as it has only two factors: 13 and 2. However, 15 is *not* prime, as it has four factors: 1, 3, 5, 15. If a number is not prime, then it is either 1 or a *composite* number. A composite number is a multiple of such number. Prime numbers also have no pattern, meaning that it is unknown how to track what the next one can be.

Every composite number can be expressed as a product of its prime factors. This is called *prime factorization*. As an example, the prime factorization of 15 is 3×5 , the prime factorization of 36 is $2^2 \times 3^2$.

Now, let us note the definition of two concepts that are important in algebra:

Definition 1.10. The greatest common factor of two or more numbers is the largest factor which is a factor of each number.

Definition 1.11. The least common multiple of two or more numbers is the smallest multiple which is a multiple of each number.

Say, the set of all factors of 24 is $\{1, 2, 3, 4, 6, 8, 12, 24\}$ and all factors of 32 is $\{1, 2, 4, 8, 16, 32\}$. The GCF of 24 and 32 is 8. Note that, we can resort to prime factorization for numbers if the numbers get too large.

As a rule of thumb: "*Finding the GCF of two or more numbers, write each number in terms of its prime factors and take the product of the different factors common to the given numbers, each taken the smallest number of times that it occurs as a factor in any of the given numbers.*"

Say 24 and 32 again. The prime factorization of 24 is 3×2^3 and 32 is 2^5 . The smallest of the 2's is 2^3 and that 3 is not a common factor in both sides, therefore we do not include it. Therefore, $2^3 = 2 \times 2 \times 2 = 8$. We will discuss more on exponents later on. But, note that we got the same GCF. Therefore, this rule works.

For LCM, say the set of multiples (up to 10) of 20 is $\{20, 40, 60, 80, 100, 120, 140, 160, 180, 200\}$, and for 25 is $\{25, 50, 75, 100, 125, 150, 175, 200, 225, 250\}$. The LCM of both is 100. However, again, we can use prime factorization as an intuitive way to find the LCM of two numbers.

As another rule of thumb: "*To find the LCM of two or more numbers, write each number in terms of its prime factors and then take the product of the different factors each taken the greatest number of times that it occurs as a factor in any of the given.*"

Say 20 and 25 again. The prime factorization of 20 is 5×2^2 and 25 is 5^2 . The largest of the two's is 2^2 , and the largest of the 5's is 5^2 . Therefore, our LCM is:

$$\begin{aligned}
 &= 2^2 \times 5^2 \\
 &= 2^2 \times (5 \times 5) \\
 &= 2^2 \times 25 \\
 &= (2 \times 2) \times 25 \\
 &= 4 \times 25 \\
 &= 100
 \end{aligned}$$

Therefore 100 is our LCM. However, LCM and GCF can be found through formulas. The following can be defined:

Definition 1.12 (Formula for GCF). Let $a, b, c, \dots \in \mathbb{Z}$. To find the GCF of two or more numbers:

$$gcf(a, b, c, \dots) = \frac{a \times b \times c \times \dots}{lcm(a, b, c, \dots)}$$

Definition 1.13 (Formula for LCM). Let $a, b, c, \dots \in \mathbb{Z}$. To find the LCM of two or more numbers:

$$lcm(a, b, c, \dots) = \frac{a \times b \times c \times \dots}{gcf(a, b, c, \dots)}$$

We will not further discuss this to keep this discussion short, but examples are left for the reader.

1.3.5 Exponents and Powers

An *exponent* is a representation on how many times a certain number can be multiplied. These are also called powers, or indices. Note that, this is an introductory course related to powers. More on about powers are in Chapter 7.

Definition 1.14. Let $a \in \mathbb{R}$. Any $a^n = a \times a \times a \times \dots$, wherein a is being multiplied an n amount of times.

We consider a as the *base*, and n is the *power*. For lack of confusion, we will assume all that n is the *power* and NOT the exponential. Textbooks may confuse with such.

The following are the basic laws of exponents. We will consider them as theorems for now.

Theorem 1.19. $a^m \times a^n = a^{m+n}$

Theorem 1.20. $a \neq 0, \frac{a^m}{a^n} = a^{m-n}$

Theorem 1.21. $(a \times b \times \dots)^m = a^m \times b^m \times \dots$

Theorem 1.22. $(\frac{a}{b})^m = \frac{a^m}{b^m}$

Theorem 1.23. $(a^m)^n = a^{mn}$

We will assume that no proof will be provided, as they will be an exercise for the reader.

FINISHING REMARKS – All of these provided are necessary to proceed in algebra. However, more topics will be introduced. By the next topic, three topics; the real number line, order in the real numbers, and absolute value, are going to introduce the bane of graphing and plotting. However, this may take some time due to the sheer will to do this in L^AT_EX, therefore will need to take a time off to continue forward. In various cases, all exercises will be provided in the book(s) cited. Defining all of what is noted, we are going to be discussing these some day later. For now, exercises may first be provided for the reader before proceeding, and a time for the author to fully understand TikZ by some standpoint. Therefore, I will now depart.

Date of Beginning - 17/04/24, 21:01.

Date of End - 19/04/24, 18:44.

1.4 May 23, 2024.

[22:17] This may have been the longest gap that I have ever experienced. Essentially, I had a massive burnout to learning college algebra and decided to take a one month break. Exams, a ball, school, and some stupid things, but it is finally time to return. But overall, we will return. We will return with The Real Number Line, Concept of order, and Absolute Value. We shall begin. But before so, do note that *this will start to be transition to have personal additions, humor, or ideas to this entire note lodge. I will also include examples this time, since I wish to make sure that I get this when I eventually grow up to get a more ideal response.* After this information, let us begin:

1.4.1 The Real Number Line

The *real number line* is the line of all real numbers on its points. Essentially, every real number is on every point on the line, and every point on the line corresponds to a real number. Look at the figure below to infer, as it is important.

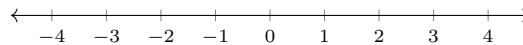


Figure 1: The Real Number Line

0 in the graph is titled the *origin*, and everything else is considered a plot. On the right of the origin, there are the *negative* real numbers. On the left of the origin, there are the *positive* real numbers. All real numbers have a plot, as well as the irrationals. Therefore, if I were to plot, say, π , it would be as follows:

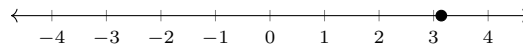


Figure 2: Plot of π

Of course, the graph is not accurate. But this is most accurate to represent the graph π . If I were to plot an ordinary number, say, 2, it would be as: A *plot* is a dot on a specific coordinate.

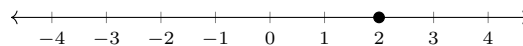


Figure 3: Graph of 2

1.4.2 Concept of Order

In terms of order, there are two symbols that define order in the real numbers. Accredited by Rene Descartes and symbolically formalized by Thomas Harriot:

Definition 1.15. $\forall x, y \in \mathbb{R}, \exists x > y \vee x < y$.

If $x > y$, then x is bigger than y . If $x < y$, then x is less than y . Other way around works. These statements are called *inequalities*. Another definition is the following.

Definition 1.16 (Inequality). $\forall a, b \in \mathbb{R}$, then:

$a < b$ if $b - a$ is positive.

$a > b$ if $b < a$.

Another kind of inequality is one that includes equality with the number, as a possibility.

Definition 1.17. $a \geq b$ if $a > b \vee a = b$. $a \leq b$ if $a < b \vee a = b$.

Both of these definitions state that they can be either. $5 \geq 4$ is true, but $5 = 4$ is not, and vice versa.

To further accompany these definitions, there are two additional axioms abruptly named the *order axioms* to showcase the following:

Axiom 1.1 (Axiom of Trichotomy). $\forall a, b \in \mathbb{R}$, then $a < b$, $a > b$, and $a = b$ cannot be true all at once.

Axiom 1.2 (Axiom of Closure for Positives). $\forall a, b \in \mathbb{R} \wedge a > 0, b > 0$, then $a + b > 0 \wedge ab > 0$.

1.4.3 Absolute Value

The definition of absolute value in layman's terms is the numerical value of the number regardless of the sign. A more symbolic approach is:

Definition 1.18. $\forall a \in \mathbb{R}$, the absolute value of a $|a|$ is:

$$|a| = a \text{ if } a \geq 0.$$

$$|a| = -a \text{ if } a < 0.$$

It is also defined as the distance from one point from zero. Distance can't be negative, so we can tell that the absolute value of a number is *always nonnegative*. $|-4| = 4$, same as $|4| = 4$, since it tells how far -4 or 4 is from the origin. Illustrated on the number line:

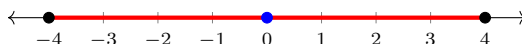


Figure 4: $|4|$ or $|-4|$ represented in the number line

Marked in red is the distance from zero -4 and 4 have. It is 4 . It is exactly 4 units from either point. To find the distance between two points on the number line:

Definition 1.19. If points P and Q exist, and their corresponding numbers are respectively a and b , their distance is always either $|a - b|$ or $|b - a|$.

If I were to find the distance between -1 and 3 is $|-1 - 3|$ or $|3 - (-1)|$ which is 4 units.

1.4.4 Interval Notation

An *interval* is a subset of $\mathbb{R}$ that only specify a specific set of numbers in between two points of numbers. Seen below:

Definition 1.20 (Intervals). As follows:

1. *Closed Intervals* are intervals that are from a to b , including a and b . Symbolically, $[a, b] = \{x | a \leq x \leq b\}$.
2. *Open Intervals* are intervals that are from a to b , excluding a and b . Symbolically, $(a, b) = \{x | a < x < b\}$.
3. *Open to Right Intervals* or *Right-Half Open Intervals* are intervals from a to b , including a but excluding b . Symbolically, $[a, b) = \{x | a \leq x < b\}$.
4. *Open to Left Intervals* or *Left-Half Open Intervals* are intervals from a to b , excluding a but including b . Symbolically, $(a, b] = \{x | a < x \leq b\}$.

On the number line, an *open circle* is used to denote exclusion (or an open interval). A *closed circle* is used to denote inclusion (or a closed interval). Seen on the two figures below:



Figure 5: Open Interval $(-2, 1)$

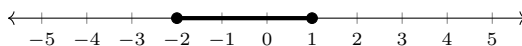


Figure 6: Closed interval $[-2, 1]$

Seen from Figure 5, we use an open circle to denote exclusion. Seen from Figure 6, we use a closed circle to denote inclusion. In their notation, every parenthesis adjacent to either/both a or b says it's excluded. Every bracket adjacent to either a or b says it's included. An *infinite interval* is an interval denoted by $(a, \pm\infty)$. Symbolically, it is a set such that $\{x | x > a\}$ exists. As represented on the number line:

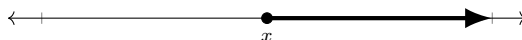


Figure 7: Infinite Interval (a, ∞)

This shows (a, ∞) which is an open interval. It can also go in the opposite direction, shown as a closed interval:

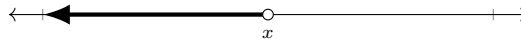


Figure 8: Infinite Interval $(-\infty, a]$

Note that the infinity does not have a closed or open interval. It just says that only one side of the line is in the interval. The interval $(-\infty, \infty)$ is the number line itself.

FINISHING REMARKS – *This entire topic is the scope on graphing on a number line dependent on solutions for inequalities later discussed in a future topic. One topic is omitted, The Completeness Axiom, but it is not really regarded as important for this stage of College Algebra, but future discussions will be introduced for terms of fields. It is also essential to learn intervals as they will be useful in mathematics. However, I must admit that this topic is not that hard to learn. It is just due to my lack or incapability of studying at the time, so when I took a month break and after it, I still did not return as I was still practicing for my solving skills in another L^AT_EX document. However, I feel like I have the motivation to continue especially after I learnt a bit of TikZ and PgfPlots in the process. This will greatly help with the creation of figures and graphs. As I depart once again, I thank that I have finally returned to creating LaTeX documents after a month-long break.*

Date of Beginning - 24/05/24, 22:17.

Date of End - 30/05/24, 00:36.

I will now bid my farewell to this chapter, as we transition to a more greater feat: Introduction of Ordered Pairs, The Cartesian Product, Relations vs. Functions, Operations on Functions, Composition, Kinds of Functions, Inverse Functions, The Cartesian-Coordinate Plane, Graphs of Relations, and Graphs of Functions.

2 The Concept of Relations and Functions

We shall now begin our journey to understanding how a relation or functions works. But firstly, let us define an ordered pair.

2.1 May 30, 2024.

[20:22] We shall now embark on our journey on the introduction on one of the most important topics in mathematics. We shall dissect them, one by one. We shall proceed:

2.1.1 Introduction of Ordered Pairs

An *ordered pair* is a pair of two elements, commonly as x and y , that have an order. They are notated as (x, y) . x is the abscissa. y is the ordinate. All ordered pairs are NOT equal. They are only equal if:

Definition 2.1 (Equal Ordered Pairs). If (a, b) and (c, d) are ordered pairs, then $(a, b) = (c, d) \Leftrightarrow a = c \wedge b = d$.

All ordered pairs are IN ORDER. Interchanging the abscissa or the ordinate with each other will change the pair. $(3, 5) \neq (5, 3)$, however $\{3, 5\} = \{5, 3\}$. Ordered pairs are different unlike subsets, which have elements that can change. Ordered pairs can be represented as plots in a graph, but we will mention that after we introduce the Cartesian Plane.

2.1.2 The Cartesian Product

The *cartesian product*, denoted by $A \times B$, wherein A and B are sets, gives the set of all possible ordered pairs that can be formed from each of the set's elements. An example: Let $A = \{2, 3\}$ and $B = \{a, b, c\}$. Therefore, $A \times B = \{(2, a), (2, b), (2, c), (3, a), (3, b), (3, c)\}$.

$A \times A$ and $B \times B$ are special cases of cartesian products. Note that, cartesian products are NOT commutative in general. Therefore, $A \times B \neq B \times A$. Their elements would be different due to the definition of an ordered pair. However, the cardinality of a cartesian product is the same of the product of each set separately. Or, $Card(A \times B) = Card(A) \times Card(B)$. The cardinality of a cartesian product is also commutative, so:

$$Card(A \times B) = Card(B \times A)$$

$$\text{or } Card(A) \times Card(B) = Card(B) \times Card(A)$$

2.1.3 Relations vs. Functions

A *relation* is the relationship between the abscissa and the ordinate. They can have one abscissa, having multiple ordinates. Any subset of $A \times B$ is considered a relation. Generally, in a relation, the abscissa is the input and the ordinate is the output.

All potential inputs of a relation is called the *domain*, and all potential outputs or the relation is called the *range*. Inputs and outputs are the abscissa and ordinates of the relation, respectively. Symbolically: Let S be a relation from sets A to B .

$$D(S) = \{x | (x, y) \in S\}$$

$$R(S) = \{y | (x, y) \in S\}$$

If there exists a set A , $A \times A$ is a relation on A , per any subset in $A \times A$. There also exists a set S of elements (ordered pairs) $(a, b) \in A \times A$. All elements not present in S , or S' , is the *complementary relation* on S .

An *inverse relation* is the relation S in $S \subseteq A \times B$, wherein $x \in A$, and $y \in B$. The inverse relation is denoted as S^{-1} . In layman's terms, it is a relation wherein the output of the original relation returns the input, or as a backward process. Seen in the following:

$$D(S) = R(S^{-1})$$

$$R(S) = D(S^{-1})$$

A *function* is a special kind of relation wherein the domain only has a representation to a specific valued range, and not to multiple values in the range. The notation of a function is $f(x) = y$. x is the domain or the abscissa, and y is the range or the ordinate. Note the following:

- Relations (a, b) , (a, c) , and (a, d) are NOT functions. They do not correspond one-to-one, as a has multiple kinds of outputs.
- Relations (a, b) , (c, d) , and (e, f) are functions. They correspond one-to-one between two values in the domain and range.

Notation $f(x) = y$ shows that: x is considered the *argument* and y is considered the *dependent variable*. Unlike relations, functions can be represented by an algebraic sentence that can only yield one output from each domain.

Definition 2.2. $\forall x \in A \wedge \forall y \in B$, wherein A and B are sets, $\exists y \in B$ that is unique $\forall x \in A$. In simple terms: For any x , every output $y \in B$ is unique to $x \in A$.

If we were to give a more proper definition on how a function is: It is like a machine. It is a machine that has its wires as an algebraic sentence. The input of the machine is the domain. The yield is the range. One output (range) can be made from multiple inputs (domains), but one input making multiple outputs is not a function. For example, $f(x) = 3x - 2$. $3x - 2$ is our algebraic sentence. We can therefore input an x such that it yields an output y that is unique to any x , and not the other way around. If we have $x = 0$:

$$\begin{aligned} f(x) &= 3x - 2 = f(0) = 3(0) - 2 \\ &= 0 - 2 \\ &= -2 \end{aligned}$$

If we have, say, $x = 1$:

$$\begin{aligned} f(x) &= 3x - 2 = f(1) = 3(1) - 2 \\ &= 3 - 2 \\ &= 1 \end{aligned}$$

Or even $x = -1$:

$$\begin{aligned} f(x) &= 3x - 2 = f(-1) = 3(-1) - 2 \\ &= -3 - 2 \\ &= -5 \end{aligned}$$

You can probably tell the gist of how functions work. It is basically giving an input, using the algebraic sentence (with the input) to make the output.

2.1.4 Operations on Functions

Functions can have operations performed on them. Functions can also co-exist to each other. They can perform arithmetic, as follows:

Let f and g be functions of $f(x)$ and $g(x)$, and x is the domain of both functions. Therefore:

Definition 2.3 (Sum of Functions). The sum of f and g is defined as:

$$(f + g)(x) = f(x) + g(x)$$

Definition 2.4 (Difference of Functions). The difference of f and g is defined as:

$$(f - g)(x) = f(x) - g(x)$$

Definition 2.5 (Product of Functions). The product of f and g is defined as:

$$(fg)(x) = f(x)g(x)$$

Definition 2.6 (Quotient of Functions). The difference of f and g is defined as:

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$

Given $g(x) \neq 0$.

Examples may be given:

Example 2.1. Let $f(x) = 4x - 5$ and $g(x) = x + 12$. Find $(f + g)(x)$.

$$\begin{aligned}(f + g)(x) &= f(x) + g(x) = 4x - 5 + x + 12 \\ (4x + x) + (-5 + 12) &= \boxed{5x + 7}\end{aligned}$$

Therefore, $(f + g)(x) = 5x + 7$.

Example 2.2. Let $f(x) = \frac{1}{2}x + 9$ and $g(x) = -3x$. Find $(f - g)(x)$.

$$\begin{aligned}(f - g)(x) &= f(x) - g(x) = \frac{1}{2}x + 9 - (-3x) \\ \frac{1}{2}x + 9 + 3x &= \left(\frac{1}{2}x + 3x\right) + 9 \\ &= \left(\frac{1}{2}x + \frac{6}{2}x\right) + 9 \\ &= \boxed{\frac{7}{2}x + 9}\end{aligned}$$

Therefore, $(f - g)(x) = \frac{7}{2}x + 9$.

Example 2.3. Let $f(x) = x - 5$ and $g(x) = 2x + 1$. Find $(fg)\left(-\frac{1}{2}\right)$.

$$\begin{aligned}(fg)(x) &= f(x)g(x) = (x - 5)(2x + 1) \\ x(2x + 1) - 5(2x + 1) &= 2x^2 + x - 10x + 5 \\ &= 2x^2 - 9x + 5\end{aligned}$$

Find $(fg)\left(-\frac{1}{2}\right)$.

$$\begin{aligned}
 (fg)\left(-\frac{1}{2}\right) &= 2\left(-\frac{1}{2}\right)^2 - 9\left(-\frac{1}{2}\right) - 5 \\
 2\left(\frac{1}{4}\right) - \left(-\frac{9}{2}\right) - 5 &= \frac{2}{4} + \frac{9}{2} - 5 \\
 &= \frac{1}{2} + \frac{9}{2} - 5 \\
 &= \frac{10}{2} - 5 \\
 &= 5 - 5 \\
 &= \boxed{0}
 \end{aligned}$$

Therefore, $(fg)\left(-\frac{1}{2}\right) = 0$.

Example 2.4. Let $f(x) = x^2 - 5$ and $g(x) = x^2 - 3x - 18$. Find $\left(\frac{f}{g}\right)(6)$.

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{x^2 - 5}{x^2 - 3x - 18}$$

Find $\left(\frac{f}{g}\right)(6)$.

$$\begin{aligned}
 \left(\frac{f}{g}\right)(6) &= \frac{(6)^2 - 5}{(6)^2 - 3(6) - 18} \\
 \frac{36 - 5}{36 - 18 - 18} &= \frac{31}{18 - 18} \\
 &= \boxed{\frac{31}{0} \text{ is undefined.}}
 \end{aligned}$$

Therefore, $\left(\frac{f}{g}\right)(6)$ is undefined in terms of that function.

2.1.5 Composition

Function composition is defined as putting a function within a function, to put simply.

Definition 2.7 (Function Composition). If there exists functions $f(x)$ and $g(x)$, their composition $(f \circ g)(x) = f(g(x))$. All domain and range applied to g must be applied to f .

Two examples can be given:

Example 2.5. If $f(x) = 2x$ and $g(x) = 3x + 7$, find $(f \circ g)(x)$.

$$\begin{aligned}
 (f \circ g)(x) &= f(g(x)) = f(3x + 7) \\
 2(3x + 7) &= \boxed{6x + 14}
 \end{aligned}$$

Therefore, $(f \circ g)(x) = 6x + 14$.

Example 2.6. If $f(x) = 3x$, and $g(x) = \frac{1}{2x-1}$, find $(f \circ g)\left(\frac{1}{2}\right)$.

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) = f\left(\frac{1}{2x-1}\right) \\ (3)\left(\frac{1}{2x-1}\right) &= \frac{3}{2x-1}\end{aligned}$$

Find $(f \circ g)\left(\frac{1}{2}\right)$.

$$\begin{aligned}(f \circ g)\left(\frac{1}{2}\right) &= \frac{3}{2\left(\frac{1}{2}\right) - 1} \\ \frac{3}{1-1} &= \boxed{\frac{3}{0} \text{ is undefined}}\end{aligned}$$

Therefore, $(f \circ g)\left(\frac{1}{2}\right)$ is undefined with this function.

Note that in Example 2.6, the domain of the function g specifies that it is undefined when $x = \frac{1}{2}$. This domain also applies to f as it is composed of it. If we were to do $g\left(\frac{1}{2}\right)$ by itself, we would end up an undefined function. Therefore, the domain and range must also be applied to f . Function composition is *not commutative*. $(f \circ g)(x) \neq (g \circ f)(x)$. As an example:

Example 2.7. Let $f(x) = x + 1$, and $g(x) = 2x$. Find out if $(f \circ g)(x) = (g \circ f)(x)$ is true. For $(f \circ g)(x)$:

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) = f(2x) \\ &= \boxed{2x + 3}\end{aligned}$$

For $(g \circ f)(x)$:

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) = g(x + 1) \\ 2(x + 1) &= \boxed{2x + 2}\end{aligned}$$

Therefore, $(f \circ g)(x) = (g \circ f)(x)$ is false.

FINISHING REMARKS – This topic involves the topic on functions, and their basis. So far, we have introduced thoroughly onto their concept, and how they can be modified to make newer functions. The next couple topics will tackle on the types of functions found, the cartesian plane, and graphing relations. This topic will elude to a more bigger and broader expanse of polynomials that will be discussed in the future. More about functions can be heard of in some future chapter, as this only serves an introduction for functions and relations in general. A few days will be omitted because the writer will be taking a short trip, and will need a rest day afterwards. Anyways, I shall now depart from this topic for this day and move on to the next.

Date of Beginning - 30/05/24, 20:22.

Date of End - 31/05/24, 17:28.

2.2 June 3, 2024.

[14:53] This subsection will talk about the types of functions, defining the cartesian plane, graphing relations, and all of the above mentioned before.

2.2.1 Domain and Range

Domains and range are used to determine inputs and outputs of functions respectively. However, functions can have a variety of outputs from various inputs¹. For our definitions, we will have to define new terms. The *codomain* are all the potential range that can be made from a domain. The *range* are all the outputs that are from the codomain. The *image* is the range. Look at the image below:

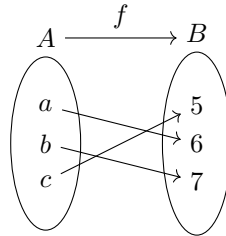


Figure 9: Function Mapping

If a function f has two sets of domains and range, A and B , then the notation of a function with domains A and B is $f : A \rightarrow B$. The following can be defined:

1. *Bijjective Function* – a function which has exact correspondence between every element in the image. Simply put, it maps every output without skipping or having the same elements in the image. $f(x) = x$ is a bijective function, as it includes all x in the image B , and maps exactly all of the codomain. (considering that all elements of domain A is all possible x .)
2. *Injective Function* – a function where every element of the domain A has a unique output in the codomain B . Every bijective function is considered injective, as they map exactly ONE unique element in the codomain.
3. *Surjective Function* – a function wherein every element of the domain A has an output in the codomain B wherein all elements of B are given a pre-image (an element in A) no matter what. They can have duplicates, but no element in B is forgotten. $f(x) = x^2$ is a surjective function. If we were to use pre-images 1 and -1 , they will both lead to 1.
4. *Many-to-one Function* – a function wherein every element in the domain A has an output in the codomain B , wherein if $a, b \in A$ and $c \in B$, then $f(a) = c$ and $f(b) = c$. In simple terms, it is when any element in the domain lead to one specific element in the image. Consider $f(x) = x^2$ again. It is also a many-to-one, as every negative input leads to a positive output, same for a positive input.

¹The following definitions are hyper-simplified. They are more expressed in Abstract Algebra which is beyond the fields that we are discussing right now.

Figure 9 is considered a bijective function, since every element in A maps to every element in B . Figure 10 (below) is considered a surjective function, as every element in A has an output in B , but does not leave anything out.

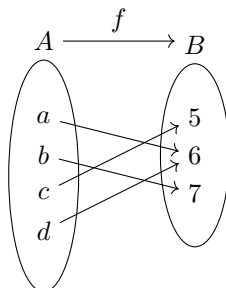


Figure 10: Surjective Function

2.2.2 Piecewise Functions

A *piecewise function* is a function that has specific cases that lead to different images.

Definition 2.8 (Piecewise Function). If $f(x) = a$ and $f(x) = b$ and c or d are cases, respectively to each function, then,

$$f(x) = \begin{cases} a & \text{if } c \\ b & \text{if } d \end{cases}$$

Basically, if x is something then the function will do this instead. However, if x is something else then the function will do something else instead. Piecewise functions can have any amount of cases for the domain to be specific to it.

Example 2.8. If $f(x) = \begin{cases} 3x & \text{if } x > 0 \\ x - 1 & \text{if } x = 0 \\ x & \text{if } x < 0 \end{cases}$, find $f(2)$, $f(0)$, and $f(-1)$.

- $f(2)$: Since $2 > 0$, $f(2) = 3(2)$, which means $\boxed{f(2) = 6}$.
- $f(0)$: Since $0 = 0$, $f(0) = 0 - 1$, which means $\boxed{f(0) = -1}$.
- $f(-1)$: Since $-1 < 0$, $\boxed{f(-1) = -1}$.

Example 2.9. If $f(x) = \begin{cases} 2x - 1 & \text{if } x > 2 \\ x + 4 & \text{if } 0 < x < 2 \end{cases}$, find $f(4)$, $f\left(\frac{3}{2}\right)$, and $f(-2)$.

- $f(4)$: Since $4 > 2$, $f(4) = 2(4) - 1$, which means $\boxed{f(x) = 7}$.
- $f\left(\frac{3}{2}\right)$: Since $0 < \frac{3}{2} < 2$, then $f\left(\frac{3}{2}\right) = \frac{3}{2} + 4$, which means $\boxed{f\left(\frac{3}{2}\right) = \frac{11}{2}}$.
- $f(-2)$: Since there is no case for $x < 0$, which $-2 < 0$, then $\boxed{f(-2) \text{ is undefined.}}$

2.2.3 Inverse Functions

An *inverse function* is a function wherein one element of the image gives back the corresponding element in the domain. In other words, the backwards process of a regular function. Finding inverse functions can be defined as the following. However, not always will an inverse function *be* a function. This will be discussed more in depth later. To find the inverse function of a function goes as follows:

Definition 2.9 (Finding an Inverse Function). If a function $f(x)$ has an inverse $f^{-1}(x)$, to find the inverse:

1. Switch all x 's to y 's, and vice versa.
2. Solve for y .
3. (optional) Test if the inverse function is true.

Let us try with three examples:

Example 2.10. If $f(x) = 5x + 9$, find $f^{-1}(x)$.

Since we know that every function $f(x) = y$, we can infer that $y = 5x + 9$ in our case.

1. Switch all x 's to y 's, and vice versa.

$$y = 5x + 9 = x = 5y + 9$$

2. Solve for y .

$$\begin{aligned} \text{(Solve for } y) \quad -5y &= -x + 9 = \frac{-5y}{-5} = \frac{-x + 9}{-5} \\ y &= \frac{-x + 9}{-5} = \frac{-x}{-5} + \frac{9}{-5} \\ &= \frac{x}{5} + \frac{-9}{5} \\ &= \boxed{\frac{x - 9}{5}} \end{aligned}$$

$$\text{Therefore, } f^{-1}(x) = \frac{x - 9}{5}.$$

3. (optional) Test if the inverse function is true.

We shall try $f(1)$ as our first test, and its output will be tested if it yields 1 back.

$$f(1) = 5(1) + 9 = 14$$

Now, let us try $f^{-1}(14)$.

$$\begin{aligned} f^{-1}(14) &= \frac{14 - 9}{5} = \frac{5}{5} \\ &= 1 \end{aligned}$$

We can confirm that the function and its inverse are true.

Example 2.11. If $f(x) = \frac{2x - 3}{7}$, find $f^{-1}(x)$. Find, and show $f^{-1}(3)$ is true to the domain of the function (such that it will return back to 3).

Since we know that every function $f(x) = y$, we can infer that $y = \frac{2x - 3}{7}$.

1. Switch all x 's to y 's, and vice versa.

$$y = \frac{2x - 3}{7} = x = \frac{2y - 3}{7}$$

2. Solve for y .

$$\begin{aligned}
 7(x) &= 7\left(\frac{2y-3}{7}\right) = 7x = 2y - 3 \\
 -2y &= -7x - 3 = \frac{-2y}{-2} = \frac{-7x-3}{-2} \\
 y &= \frac{-7x-3}{-2} = \frac{-7x}{-2} + \frac{-3}{-2} \\
 &= \frac{7x}{2} + \frac{3}{2} \\
 &= \boxed{\frac{7x+3}{2}}
 \end{aligned}$$

Therefore, $f^{-1}(x) = \frac{7x+3}{2}$. Now, we need to find $f^{-1}(3)$.

$$\begin{aligned}
 f^{-1}(3) &= \frac{7(3)+3}{2} = \frac{21+3}{2} \\
 &= \frac{24}{2} \\
 &= 12
 \end{aligned}$$

Now, let us verify this by putting 12 back to the original function.

$$\begin{aligned}
 f(12) &= \frac{2(12)-3}{7} = \frac{24-3}{7} \\
 &= \frac{21}{7} \\
 &= 3
 \end{aligned}$$

We can confirm that the function and its inverse are true.

Example 2.12. If $f^{-1}(x) = \frac{3x+1}{5}$, find $f(x)$. Evaluate $f(2)$.

We can infer that (most) inverse functions act as regular functions but just work backwards. So, infer that we can use the same steps to find the original function.

1. Switch all x 's to y 's, and vice versa.

$$y = \frac{3x+1}{5} = x = \frac{3y+1}{5}$$

2. Solve for y .

$$\begin{aligned}
 5(x) &= 5\left(\frac{3y+1}{5}\right) = 5x = 3y + 1 \\
 -3y &= -5x + 1 = \frac{-3y}{-3} = \frac{-5x+1}{-3} \\
 y &= \frac{-5x+1}{-3} = \frac{-5x}{-3} + \frac{1}{-3} \\
 &= \frac{5x}{3} + \frac{-1}{3} \\
 &= \boxed{\frac{5x-1}{3}}
 \end{aligned}$$

Therefore, $f(x) = \frac{5x-1}{3}$. Now, we need to find $f(2)$.

$$\begin{aligned} f(2) &= \frac{5(2)-1}{3} = \frac{10-1}{3} \\ &= \frac{9}{3} \\ &= 3 \end{aligned}$$

Verify this by putting 3 back the inverse function:

$$\begin{aligned} f^{-1}(3) &= \frac{3(3)+1}{5} = \frac{9+1}{5} \\ &= \frac{10}{5} \\ &= 2 \end{aligned}$$

We can confirm that the function and its inverse are true.

I have previously mentioned that: All functions have inverses, but not all inverses are functions.

Example 2.13 (Invalidity of Some Inverse Functions). Let a function $f : A \rightarrow B$:

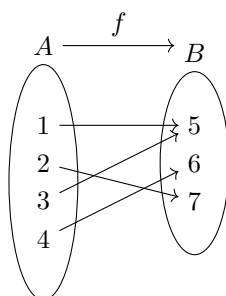


Figure 11: Function Diagram

This is a surjective function f . If we were to get its inverse:

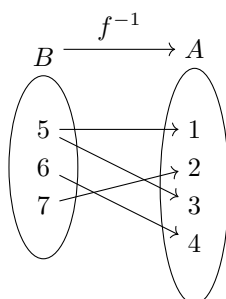


Figure 12: Inverse of f

The original set of relations $\{(1, 5), (2, 7), (3, 5), (4, 7)\}$ are in the form of a surjective function. However, the inverse function's set of relations $\{(5, 1), (5, 2), (6, 3), (6, 4), (7, 2)\}$ are not a function. One element of the domain, 5 (originally the range of f) gives two outputs (originally the domain of f). This cannot happen. Therefore, the inverse function does not consider itself as a function.

2.2.4 The Cartesian Plane

The *cartesian plane*, or the *coordinate plane* is a plane of two number lines forming a perpendicular intersection to make a graph. See the figure below.

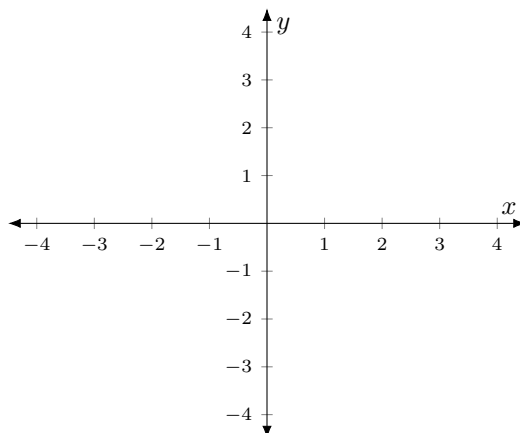


Figure 13: The Cartesian Plane

*Ignore the scale of the figure, that's a $\text{\LaTeX}$ issue for future me to fix.

Anyway, as seen on the figure, the two lines divide to four quadrants. The vertical line is called the *y-axis* or the *ordinate*. The horizontal line is called the *x-axis* or *abscissa*. Every quadrant has their own signs. This is listed counterclockwise from the top right:

1. QUADRANT I: Their signs are $(+,+)$.
2. QUADRANT II: Their signs are $(-,+)$.
3. QUADRANT III: Their signs are $(-,-)$.
4. QUADRANT IV: Their signs are $(+,-)$.

Notice that we are labelling them as relations? We can infer that a relation has an x value (domain) and a y value (range). These are specified coordinates on the graph. This is why we label the lines the y -axis and x -axis respectively. Every y value goes up or down. Every x value goes left or right.

2.2.5 Plotting and Graphing Relations

Relations have specific coordinates on the graph. They can be plotted, and they are represented with a dot or point. However, we shall represent all the points on the graph within this interval². But note that the cartesian plane is infinite. You can add as many numbers that you want to view on the graph. However, we will not perform such because it will make weird proportions.

Example 2.14. On the cartesian plane, graph $(3,1)$ and $(-2,2)$. See it on the next page.

²We will use points within the range in the graph. For example, $(5,1)$ or $(1,5)$ is not allowed as it is outside what is seen on the plane.

Seen in the following:

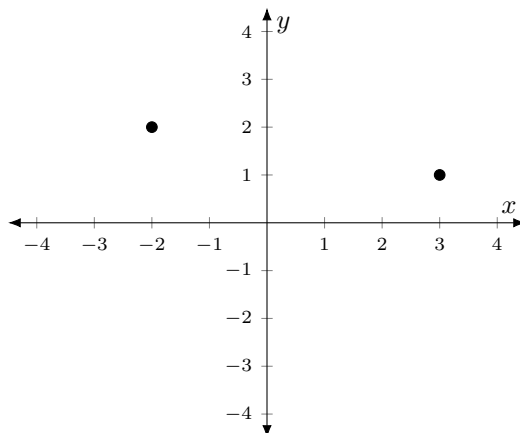


Figure 14: Plots of $(3,1)$ and $(-2,2)$

Count how many steps forward on the x -axis. If it is negative, move backwards. Then, count how many steps up on the y -axis. If it is negative, move down. You can do the y -axis first then the x -axis as well.

Example 2.15. Consider the plots in Figure 14. Draw a line to connect those points.

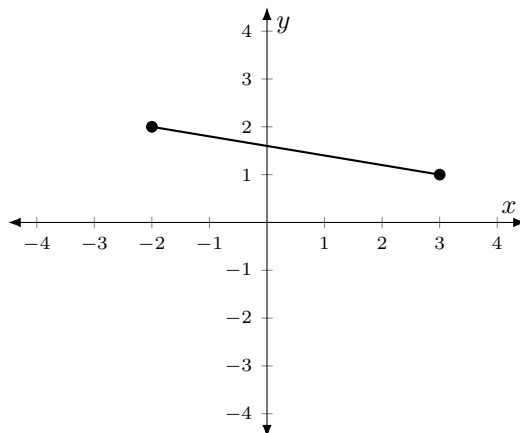


Figure 15: Plots of $(3,1)$ and $(-2,2)$

This line is the *graph*. A graph is a line that connects two points in the shortest distance and also shows all possible coordinates in the line. Every point in the plane always determines a coordinate, and its converse is true as well.

2.2.6 Plotting and Graphing Functions

A function has an element in the domain which is always represented as x , and it always has an element in the range which is always represented as y . This is important as it correlates to how the coordinate plane has an x -axis and a y -axis. Therefore, our input is the coordinate's x -axis and the output is the coordinate's y -axis.

Example 2.16. If $f(x) = x + 2$, find its graph on the cartesian plane.

1. Select sample values for our domain. Common practice is to use five. For now, we will just use three. We will use -1, 0, and 1.

2. Make a table using these sample values for our domain, and input them to our function f . These sample values are our x values, and the output is the y .

x	y
-1	1
0	2
1	3

3. Plot these points on the cartesian plane.

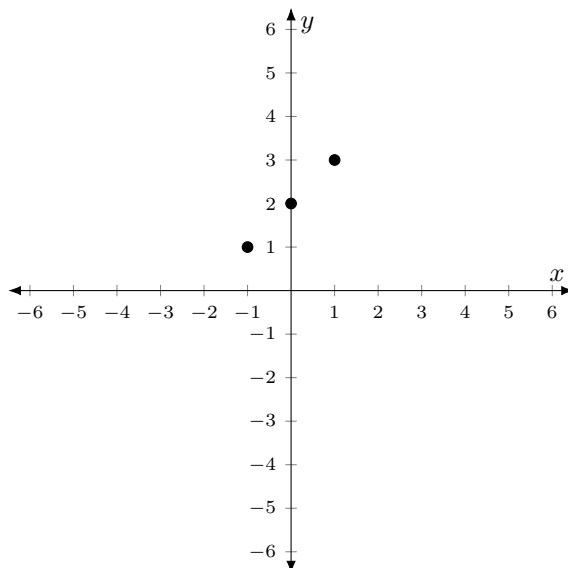


Figure 16: Plots of $(-1,1)$, $(0,2)$, and $(1,3)$

4. Connect the three points, and make a line connecting all three.

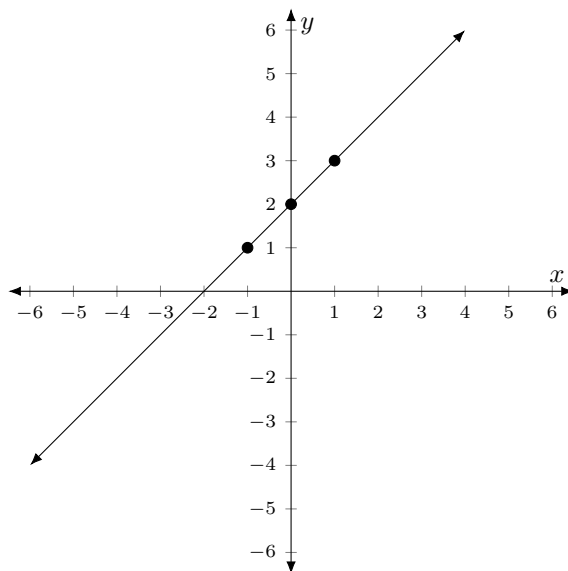


Figure 17: Graph of $f(x) = x + 2$

We extended the line so we can justify that this is an actual graph.

Therefore, this shows the graphs of $f(x) = x + 2$ in the cartesian plane. This works on most functions, however this will be discussed further more in the near future chapter.

2.2.7 Difference Between Graphs of Relations and Functions

Relations and functions are always different. However, not all graphs are functions. We can test this by the following.

Definition 2.10 (Vertical Line Test). If an imaginary line does not hit two points on any location on the graph of a relation, then it is a function.

We can try this with the graph of the common function $f(x) = x$. The process graphing it is left as an exercise for the reader.

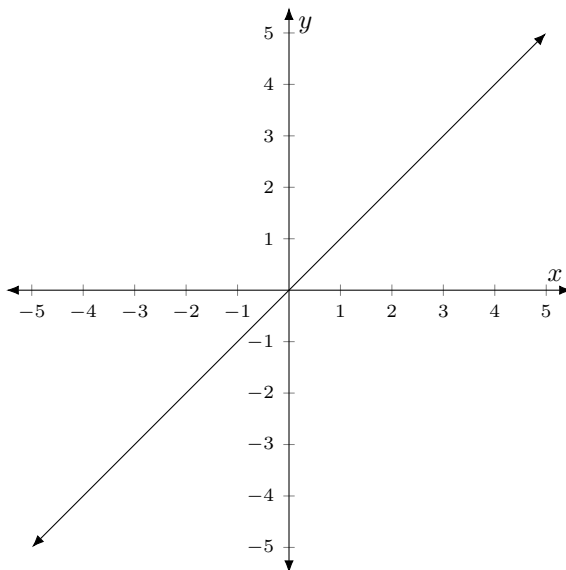


Figure 18: Graph of $f(x) = x$

Let us draw a vertical line at any point. It can be anywhere. It states "any location" for a reason.

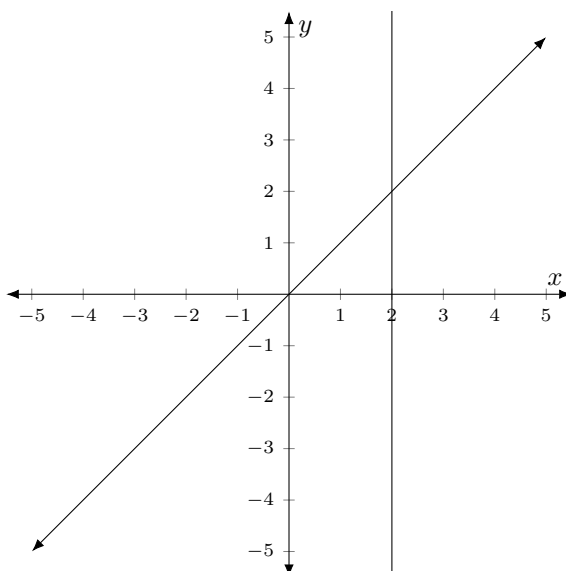


Figure 19: Vertical Line Test of $f(x) = x$

We can assure that $f(x) = x$ is indeed a function since it passes the vertical line test.

The vertical line test fails if the vertical line passes through at a place with two or more points. It passes if there are no points or consistently has one point anywhere on the graph. Let us look at an example which fails the vertical line test.

Example 2.17 (Failing the Vertical Line Test). Let a relation exist wherein $x = y^2$. We will not use functional notation for this testing for to forsee if the result is not failure or if it passes. Shown on the graph:

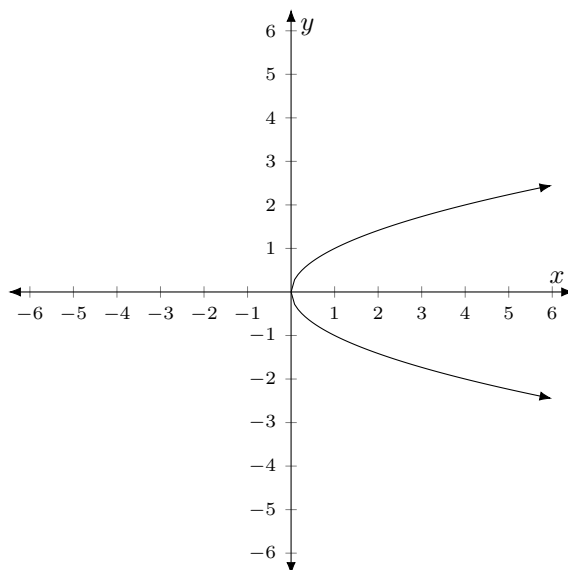


Figure 20: (inaccurate) Graph of $x = y^2$

If we were to do the vertical line test of this graph:

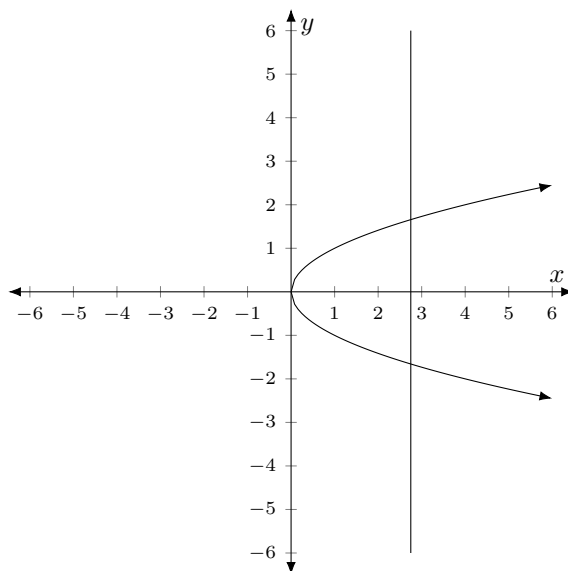


Figure 21: Vertical Line Test of $x = y^2$

As you can see, the line passes through two points of the line. Therefore, the graph fails the vertical line test, hence it is considered a relation and not a function. Although it is inaccurate³, it still

³I had to use graphs of $\sqrt{x}$ and $-\sqrt{x}$ because PGFPlots does not allow graphs of $x = y^2$. Improvisation works.

shows the relevance of what is being said.

2.2.8 Involution

Function compositions are vital to showing if two functions are each inverses of each. The process of *involution* involves the function composition of two related inverse functions.

Definition 2.11 (Involution). The following:

1. Show that $(f \circ g)(x) = (g \circ f)(x)$.
2. If $f^{-1} = g$ and $g^{-1} = f$, then both functions are related inverses.

However, if $f^{-1} \neq g$ and $g^{-1} \neq f$, then both functions are not related inverses.

The following examples showcase one with two functions as related inverses, and one that are not.

Example 2.18. $f(x) = \frac{x}{3} - 1$, and $g(x) = 3x + 3$. Show that $(f \circ g)(x) = (g \circ f)(x)$ is true, unless otherwise.

$$\begin{aligned}(f \circ g)(x) &= \frac{3x + 3}{3} - 1 \\ \frac{3x}{3} + \frac{3}{3} - 1 &= x + 1 - 1 \\ &= \boxed{x}\end{aligned}$$

Therefore, f and g are inverses of each other. If they yield back x then they are inverses of each other. This information is sufficient enough to show that each function are related inverses, but to assure we can do $(g \circ f)$.

$$\begin{aligned}(g \circ f)(x) &= 3\left(\frac{x}{3} - 1\right) + 3 \\ 3\left(\frac{x}{3}\right) - 3(1) + 3 &= x - 3 + 3 \\ &= \boxed{x}\end{aligned}$$

Example 2.19 (Failure of Involution). $f(x) = \frac{x+6}{2}$, and $g(x) = 2x + 6$. Show that $(f \circ g)(x) = (g \circ f)(x)$ is true, unless otherwise.

$$\begin{aligned}(f \circ g)(x) &= \frac{(2x + 6) + 6}{2} \\ \frac{2x + 12}{2} &= \frac{2x}{2} + \frac{12}{2} \\ &= \boxed{x + 6}\end{aligned}$$

We can conclude that f and g are not inverses of each other. Verifying it false as well with $(g \circ f)(x)$ is left as an exercise.

2.2.9 Finite Differences

The *finite difference* of a function considers the change of a function by changes in the domain x . This is more thoroughly taught in calculus, but a standard introduction can be a notion applicable.

Definition 2.12 (Finite Difference). For a given number h , and $\forall x$, the finite difference of a function f is:

$$\Delta f(x) = f(x + h) - f(x)$$

wherein $f(x + h)$ is considered the difference of $f(x)$. Note that, $\Delta f(x)$ is not a product.

h has an arbitrary value typical to 1, but any value can be assigned. We shall try some examples:

Example 2.20. Let $f(x) = 4x + 1$. Assuming $h = 3$, find $\Delta f(x)$.

$$\begin{aligned}\Delta f(x) &= f(x + 3) - f(x) \\ 4(x + 3) + 1 - 4x + 1 &= 4x + 12 - 4x + 1 \\ (4x - 4x) + 12 + 1 &= 0 + 12 + 1 \\ &= \boxed{13}\end{aligned}$$

Therefore, $\Delta f(x)$ wherein $h = 3$ is 13.

Example 2.21. Let $f(x) = x^2 + 3$. Assuming $h = -1$, find $\Delta f(x)$.

$$\begin{aligned}\Delta f(x) &= f(x + (-1)) - f(x) \\ &= f(x - 1) - f(x) \\ (x - 1)^2 + 3 - x^2 + 3 &= x^2 - 2x + 1 + 3 - x^2 + 3 \\ (x^2 - x^2) - 2x + (1 + 3 + 3) &= 0 - 2x + 7 \\ &= \boxed{-2x + 7}\end{aligned}$$

Therefore, $\Delta f(x)$ wherein $h = 3$ is $-2x + 7$. The process on how to square $(x + 1)^2$ will be shown in a future chapter.

2.2.10 Functions In Multiple Variables

Functions in two variables act the same algebraically. Graphing them is beyond our bounds on what we can discuss, but for our sakes, we will only understand them as in the basics rather than learning the graphical part. Finding their domain and range will not be included.

Definition 2.13. If $f(x, y, \dots)$ exists, and z is our range, then $f(x, y, \dots) = z$ exists.

We will do two examples with two and three variables respectively, to just give a basic understanding on what it means.

Example 2.22. Let $f(x, y) = 4x - \frac{1}{2}y$. Find $f(-1, 6)$.

$$\begin{aligned}f(-1, 6) &= 4(-1) - \frac{1}{2}(6) \\ -4 - \frac{6}{2} &= -4 - 3 \\ &= \boxed{-7}\end{aligned}$$

Therefore, $f(-1, 6) = -7$.

We shall proceed with our three-variable function:

Example 2.23. Let $f(x, y, z) = x^2 + y - 8z$. Find $f(5, 17, 3)$.

$$\begin{aligned}f(5, 17, 3) &= (5)^2 + 17 - 8(3) \\ 25 + 17 - 24 &= 42 - 24 \\ &= \boxed{18}\end{aligned}$$

Therefore, $f(5, 17, 3) = 18$.

Of course, this does not cover all of multiple variable functions. But this is a good overview if you want to wonder.

FINISHING REMARKS – *This entire scope on functions is the longest time I have spent on summarizing things in general. However, now with that accomplished, we shall overview on the importance of functions. Functions serve as an elementary topic in various fields of math, especially algebra. It is important to know how these works. More about functions will be discussed later on, but for now, this entire chapter serves as the basis of understanding how a function generally is. Including this chapter is graphing. It is essential for mathematics, and stems algebra at this level. But now, I will bid my "au revoir" on this chapter. The next chapter will involve polynomials. Polynomials are a massive enterprise for mathematics, alongside functions, graphing, etc. The next topics will involve the basis of a polynomial, the arithmetic on polynomials, kinds of polynomials, and much more that will arrive shortly before we transition to such. Functions and polynomials will be included in the future, but for now, this will be all that is mentioned. I will now bid a massive farewell, and I will now departure to move to polynomials.*

Date of Beginning - 03/06/24, 14:53.

Date of End - 08/06/24, 21:37.

3 Structures of Algebra

[13:54] This new section will now embrace on the concept of a polynomial, and their properties. This is the basis of algebra, and where x is more formalized. This will be explained later on as you might not get it yet (unless you do inference on how often x is used in the former pages.)

3.1 June 10, 2024.

This will cover the concept of a polynomial, difference between an equation and expression, domain and range of a polynomial, and their degree. This will also include operations on polynomials, which is hefty to grasp. We shall proceed, and introduce preliminaries to refresh the reader with concepts that we have introduced.

3.1.1 Concept of a Mathematical Sentence

In a mathematical sentence, there are always elements which define relation between two concepts or facts that ensure that it is true. There are two kinds.

An *equation* is a mathematical sentence that involves result. " $3+2$ " is not an equation, but " $3+2=5$ " is. An *expression* is a mathematical sentence that involves only procedure. " $3+2$ " is an expression as it directly involves the procedure but does not resolve outcome.

As a mathematical sentence they can have unknown elements. This is a pinpoint of algebra, and is seen behavior in previous sections precedingly. And, they can be represented as real language. "A value added by 2" or "A numerical value yields the product of 5 and 3." tells an expression and equation respectively wherein they have unknown items. It is ordinary to refer to them in notation as a letter.

$$3 + x = 7 \rightarrow \text{A number plus 3 is 7.}$$

$$4x \rightarrow \text{The product of 4 and a number.}$$

Choosing letters is up to the writer. The most commonly associated is x , y , and z . They can have subscripts or accents to denote difference among similar concepts or other definitions.

- i Subscripts such as $a_0, a_1, a_2, \dots$ are read as " a sub-null⁴," " a sub-one," " a sub-two" etc.
- ii. Accents such as $a', a'', a''', \dots$ are read as " a prime," " a second," " a third" etc.

The "*simplest form*" has likely been mentioned in this document multiple times, but in definition, it is the form of a numerical element that has its form that cannot be simplified or reduced more.

$$\frac{3a}{b} \text{ is in its simplest form.}$$

$$\frac{2a}{-4} \text{ is not in its simplest form, its simplest is } -\frac{a}{2}.$$

$$\frac{16a}{4a} \text{ is not in its simplest form, its simplest is 4.}$$

$$\frac{15}{7} \text{ is in its simplest form.}$$

$$\frac{a^2b^3}{ab^2} \text{ is not in its simplest form, its simplest is } ab.$$

Simplest forms are going to be more taught in simplifying radicals and simplifying fractions in a later subsubsection.

⁴For the Englishmen, it is " a sub-naught" prominently, and for the noble folk, it is " a sub-zero" prominently.

3.1.2 Fundamentals of Reckoning

Reckoning involves the four basic operations, exponentiation and radicals⁵. Most of this will involve redefining previous said in a short term, and not involve the hefty involvement of axioms for now.

1. The expression $a + b = c$ shows *addition*. c is the sum, and a and b are addends.
2. The expression $a - b = c$ shows *subtraction*. c is the difference, a is the subtrahend, and b is the minuend. Subtraction is the opposite of addition. We must also note that, $(a + b) - b = a$ as $b - b = b + (-b) = 0$.
3. The standardized expression of *multiplication* is either $a \times b$, $(a)(b)$, or ab . It is more defined to use ab for general expressions and $(a)(b)$ for expressing solutions. In $ab = c$, c is the product, a is the multiplicand, and b is the multiplier.
4. The standardized expression of *division* is either $a \div b$, $b|a^6$, a/b , or $\frac{a}{b}$. It is more defined to use $\frac{a}{b}$ for division and rational numbers alike, and $b|a$ in number theory or other applications. It is the inverse of multiplication. In $\frac{a}{b}$, c is the quotient, a is the dividend, and b is the divisor (or divider for more simpler terms.)
5. *Exponentiation* is repeated multiplication of the same number. In older times, it is labelled as involution. The expression a^b , a is the base and b is the power OR the exponential. Again, we will call b the power and not an exponential.
6. Expressed as $\sqrt[b]{a}$, the *radical* or *root* is the inverse of exponentiation. It was formerly most commonly known as involution⁷ The *principal root* is the square root, $\sqrt[2]{a}$, or $\sqrt{a}$. The principal root of a if it is positive, and b is odd, is $-a = -\sqrt[b]{a}$. Read Footnote 5 at the very bottom to know the future discussion about radicals.

3.1.3 An Addendum of Symbols Used

1. Originally called aggregation, *grouping* involves sorting elements in a sentence. Symbolically, the parenthesis $()$, bracket $[]$, brace $\{\}$ are used. Historically, the vinculum was used and is still used by other countries, represented as $\overline{ab}$.
2. $\pm$ reads as "plus or minus," and $\mp$ reads as "minus or plus."

In $a \pm b \mp c$, two cases are identified by which operation goes first, either:

$$a + b - c \text{ and } a - b + c$$

are given. HINT: Reading the symbols, whatever operation goes first is the first case e.g "plus or minus," plus goes first.

3. $\therefore$ stands for "therefore." e.g $3x = 9 \therefore x = 3$.
4. $\because$ stands for "because" or "since." e.g $x = 0 \because 3x = 0$.
5. Ellipses ...are used for an unending amount of terms in a sequence or sentence.
6. $\neq$ means "not equal" or "inequal."
7. $\lesseqgtr$ means "less or greater than."
8. Sometimes, $\geq$ and $\leq$ are used instead of $\geq$ and $\leq$ normally. **We will use the slanted version for more formality and aesthetic, because why not.**
9. $\nlessgtr$ and $\nlessgtr$ symbolise "not greater than" and "not less than" respectively.
10. $\equiv$ tell "is identical to" or "equivalent."

⁵This will be more educated in a thoroughly more thought manner in a future topic.

⁶Not to be confused with set-builder notation.

⁷Involution, nowadays, is defined as the process of the composition of two related inverses. See Part 2.2.8.

3.1.4 Constants, Variables, and Coefficients

In terms of expressions, there are various kinds of elements found in an expression. The first one is the *constant*. It serves as a regular number, with no affixed variable. For example:

1. In $3x - 13$, -13 is our constant. The reason being is that it is lone and does not affect the missing value.
2. In $5ab$, there may appear as no constant. This is an example of an expression with no constant. 0 is not a constant as it can be applied to any expression and result to the same expression.
3. $7s - 13 + 4$ has the constant -9 . Reason being is to add all constants, and you get -9 by adding -13 and 4 .

A *variable* is a letter or symbol that denotes a missing value. In our previous examples, our variables were x , a and b , and s . They can also be Greek or Hebrew letters, which is more documented in further cases.

A *coefficient* directly modifies the variable. It is always seen as the product of a coefficient and a variable. Unlike a constant, it does not modify it indirectly with respect to position⁸. The coefficient directly modifies the variable by multiplying it. Seen in our previous examples, 3 , 5 , and 7 are coefficients of the variables mentioned above.

1. $8x^2 + 11$ has the constant 11 , the coefficient 8 , and the variable x .
2. x has no constant, the coefficient is 1 , and the variable x . Reason why the coefficient is 1 because we consider that as it directly modifies the variable in some way.
3. $x + 2y + 19$ has the constant 19 , the coefficients 1 and 2 , and the variables x and y .

If there is a variable and (if there is) a coefficient, or a lone constant, they are called a *term*. In 1 . $8x^2$ is a term, so is 11 . x is also a term in 2 . because it is considered itself as an expression too.

3.1.5 Defining the Polynomial

A *polynomial* is an expression/equation that involves constants, variables, and coefficients, but does not involve division of a variable, negative exponents⁹, exponents with fractions, or a variable under a radical. For example: $\frac{1}{4}x + 11$ is a polynomial, however $\frac{7}{x}$ is not. In general, polynomials are considered *integral expressions* because no variable is being divided into. Otherwise, it is a *fractional expression*—they do divide into a variable—and not considered a polynomial.

If *at all* no variable is within a radical, it is considered a *rational expression*. However, if so, then it is an *irrational expression*. $\sqrt{11x} - 21$ is a rational expression, however $\sqrt{x} + 1$ is an irrational expression.

Arranging polynomials is commonly with descending order. Highest exponent down to the smallest. If a polynomial has one term, it is a *monomial*, two terms is a *binomial*, and three terms is *trinomial*. Four terms are typically just called *multinomials* or polynomials in general. This will be more explained in the following:

3.1.6 Degree of a Polynomial

The *degree of a polynomial* is the highest possible exponent that exists in the polynomial.

Definition 3.1 (Degree of a Polynomial). If a_n is a coefficient, x is a variable, and m is a power, then the general polynomial $a_n x^m + a_{n-1} x^{m-1} + a_{n-2} x^{m-2} + \cdots + a_2 x^2 + a_1 x + a_0 + 1$ exists. Then, m is the *degree* of the polynomial IF it is single variable.

⁸This is more clear when in the future for Transformations in Graphs.

⁹More clearer in Laws of Exponents.

Example 3.1. The following:

1. If we have a polynomial $x^3 - 11x + 14$, the degree is 3 because the highest power exists in the term x^3 .
2. If we have a polynomial of $x + 4$, the degree is 1. This is due to the fact that x has an exponent of 1 even if it does not appear so. This will be more thoroughly explained with defining Laws of Exponents and Radicals in a future lesson.
3. If we have a polynomial 3, the degree is 0. We assume that there always exists an x in the expression, and $x^0 = 1$ as a reason¹⁰ hence it has a degree of 0.
4. $x^{-3} + x^2 - 3$ is not a polynomial. There is no degree.

If there are multiple variables, the degree of the polynomial is impacted by the sum of the highest degrees of each variable lone.

Example 3.2 (Multiple Variables). In $xy^3 - 11x + x^2y - 3y^2 + 12xy - z$, the highest degree involves the sum of each of the highest degree in each variable. The highest degree in x is 2, in y is 3, in z is 1. Then $2 + 3 + 1 = 6$. Hence the degree of the polynomial is 6.

A polynomial is *homogeneous* if all of the variables have the same degree. In $3x^7 - 11x^5 + x^2y + x + y + 9y^4 - 2y^7$, it is homogeneous as the degree of x is 7 and y is 7.

3.1.7 Domain and Range of Polynomials

Another property that polynomials hold is that they accept all real values.

Definition 3.2 (Domain and Range of Polynomials). In the polynomial,

$$y = a_n x^m + a_{n-1} x^{m-1} + a_{n-2} x^{m-2} + \cdots + a_2 x^2 + a_1 x + a_0 + 1$$

noting that a_n is a coefficient, x is a variable, and m is a power. $x \in \mathbb{R} \wedge y \in \mathbb{R}$ no matter what, unless the function does not consider itself a polynomial.

We can show an example:

Example 3.3. If we have a polynomial $3x^4 + x^2 - 10$, all values exist that can make this exist. We can try with $\sqrt{2}$, and -3 . For $\sqrt{2}$, note that the solution might seem obsolete with no knowledge. Again, the reasoning is for a future topic.

$$\begin{aligned} 3(\sqrt{2})^4 + (\sqrt{2})^2 - 10 &= 3(4) + 2 - 10 \\ 12 + 2 - 10 &= 14 - 10 \\ &= -4 \end{aligned}$$

For -3 :

$$\begin{aligned} 3-3^4 + (3)^2 - 10 &= 3(81) + 9 - 10 \\ 243 + 9 - 10 &= 252 - 10 \\ &= 242 \end{aligned}$$

Since we tried it with an irrational, and an integer and rational number, we can conclude. 0 is also included, but it is left as an exercise to see if it is undefined.

3.1.8 Like and Unlike Terms

Terms that are *unlike* are terms that do not have the same structure in terms of the product of the variables. xy and $2y$ are unlike. $x + y$ and $2xy - 3y^2$ are unlike because the structure of the variables (not coefficients) are not the same. *Like* terms are terms that have the same structure in terms of the product of the variables. xy and $3yx$ are like terms because $3yx = 3(xy)$ by the axioms bound in the real numbers. $2x$ and x are like terms because they have the same structure. This is important for operations on polynomials. Speaking of which ...

¹⁰More understood in the lesson of Laws of Exponents, Laws of Radicals etc. They are from a future topic.

3.1.9 Operations on Polynomials

Polynomials can have operations performed on, specifically on all four basic operations: addition, subtraction, multiplication, and division. We can continue with this in knowledge with the headers dividing the kind of operation. Shown on the following:

Addition/Subtraction on Polynomials

In adding polynomials, all the like terms are added/subtracted together. However, the sign in front of the term matters. If the term has a negative in front, then the term must include that. We can sort this by ease by assorting them to parentheses by *Associativity* and use *Commutativity* to make the polynomial manageable to the bare. Look at the following example:

Example 3.4. Add all the like terms in $3x - 2xy + 17y + 4xy - 17x + 4 - 8 + z^2$.

$$3x - 2xy + 17y + 4xy - 17x + 4 - 8 + z^2 = (3x + (-17x)) + (-2xy + 4xy) + (4 + (-8)) + 17y + z^2 \quad (1)$$

$$(-14x) + (2xy) + (-4) + 17y + z^2 = -14x + 2xy - 4 + 17y + z^2 \quad (2)$$

$$= z^2 + 2xy - 14x - 17y - 4 \quad (3)$$

For (3), I rearranged the terms to be from descending order. From highest power term to a two-variable term, to single variable, to the constant. If there is a three-variable term then it goes before the two-variable term and etc.

Rather than dealing with single-terms, we can add entire polynomials. Monomial to trinomial, binomial to binomial, etc.

Definition 3.3 (Adding/Subtracting Polynomials). If a general polynomial P_0 and another polynomial P_1 exists, $(P_0) + (P_1) = P_0 + P_1$ for if there are like terms, they can be grouped more easily.

Example 3.5. Solve $(3x^2y^2 - 4xy + 6y - x) + (4y^2x^2 - yx + y + 2x - 7)$.

$$(3x^2y^2 - 4xy + 6y - x) + (4y^2x^2 - yx + y + 2x - 7) = \text{next line because of formatting} \quad (4)$$

$$3x^2y^2 - 4xy + 6y - x + 4y^2x^2 - yx + y + 2x - 7 = 7x^2y^2 + 7y + (-5xy) + 2x - 7 \quad (5)$$

$$= 7x^2y^2 - 5xy + 7y + 2x - 7 \quad (6)$$

As seen on (5), removal of parentheses is required. On the second step in (6), using *Commutativity* is a must, and using the Axiom of *Associativity* was utilized but was omitted. Simplification leads to descending order using the ability of *Commutativity*.

Example 3.6. Solve $(3xy + 12x - y + 1) - (-12xy - x - 9y + 16)$.

$$(3xy + 12x - y + 1) - (-12xy - x - 9y + 16) = (3xy + 12x - y + 1) + (-(-12xy - x - 9y + 16)) \quad (7)$$

$$= (3xy + 12x - y + 1) + (12xy + x + 9y - 16) \quad (8)$$

$$3xy + 12x - y + 1 + 12xy + x + 9y - 16 = 15xy + 13x + 8y - 15 \quad (9)$$

For (7), we used the Definition of Subtraction. For (8), we distributed the negative sign. Since we know by **Theorems 1.13 & 1.14**, and by also applying **Theorems 1.7**, the idea holds true, and is required for subtracting polynomials. These theorems will be needed to apply to Division on Polynomials.

Multiplication on Polynomials

Multiplying monomials is as simple as combining all terms into one. If there is an exponent with the base as the same variable with another one, combine them (using **Theorem 1.19**). This will be more clearer in the future topic about exponents and radicals. See the example:

Example 3.7. Solve $(4xy^3)(9x^2y)$.

$$(4xy^3)(9x^2y^2) = (4 \times 9)(x \times x^2)(y^3 \times y^2) \quad (10)$$

$$36 \times x^3 \times y^5 = 36x^3y^5 \quad (11)$$

$$\boxed{\therefore (4xy^3)(9x^2y) = 36x^3y^5}.$$

As seen in (7), we use *Associativity* to arrange the terms with the same variable product. Matter for like terms is not needed in multiplication of polynomials. *Distributivity* is commonly used, and so the following idea is shown when performing such with a degree one polynomial as the multiplier.

Example 3.8. Solve $-(11x - 9)$.

$$-3(11x - 9) = -3(-11x) + (-3(-9)) \quad (12)$$

$$= 33x + 27 \quad (13)$$

$$\therefore -3(11x - 9) = 33x + 27$$

If we were to solve with general polynomials (binomials and trinomials), the usage of *Distributivity* is necessary.

Example 3.9. Solve $(3x^2 - y)(x^4 + 5y - 12z)$.

$$(3x^2 - y)(x^4 + 5y - 12z) = 3x^2(x^4 + 5y - 12z) + (-y)(x^4 + 5y - 12z) \quad (14)$$

$$3x^6 + (3x^2)(5y) + (3x^2)(-12z) + (-xy^4) - 5y^2 + 12yz = 3x^6 + 15x^2y - 36x^2z - xy^4 - 5y^2 + 12yz \quad (15)$$

$$= 3x^6 - xy^4 - 36x^2z + 15x^2y - 5y^2 + 12yz \quad (16)$$

$$\therefore (3x^2 - y)(x^4 + 5y - 12z) = 3x^6 - xy^4 - 36x^2z + 15x^2y - 5y^2 + 12yz.$$

In terms of performing *Distributivity* with multiple terms, split the terms each with their corresponding signs, and make them a product of the multiplier. This will allow distributivity for polynomials to exist when multiplying them.

Division on Polynomials

The intricate idea of dividing by a polynomial is confusing for most people when first approaching this topic. Let us showcase this by doing an example:

Example 3.10. Solve $\frac{3x^2 + 34x - 24}{3x - 2}$.

Let us view it by the solution and explaining it step-by-step.

$$\begin{array}{r} x+12 \\ \underline{ 3x^2+34x-24} \\ -3x^2+2x \\ \underline{ -3x^2+2x} \\ 36x-24 \\ \underline{ -36x+24} \\ 0 \end{array}$$

We will assume the reader knows the pre-requisite of long division. Ignoring the formatting, we can deduct on the process:

We start by dividing the first term of the polynomial in the dividend $3x^2$, and dividing it to $3x^2$. By the process of simplification, we get x . Multiply x to $3x - 2$ and by using *Distributivity*, we subtract $3x^2 + 2x$. We must note however to distribute the negative sign; we need to do so to reduce/add the proceeding terms correctly. The procedure shown above though, omits the process and distributes it right away.

After such, divide the first term of the new dividend $36x$ to $3x$. By process of simplification, we get 12. Multiply 12 to $3x - 2$ and by using *Distributivity*, we get $36x + 24$. Distribute the negative sign and subtract/add any terms. We then get 0, meaning that the quotient is $x + 12$. Even if it is

a long paragraph, it is needed to understand the process.

$$\therefore \frac{3x^2 + 34x - 24}{3x - 2} = x + 12$$

There are various methods to dividing and multiplying polynomials. This will be seen later on, mostly on the section dedicated more about polynomials.

FINISHING REMARKS – *The basis on polynomials is essential for the study of math. Various scopes exist solely to deduct on how polynomials work or their applications. This is a field that is heavily reliant, and various more is indeed. This topic introduces the bare introductory showing of polynomials, and the next will involve a more abstract definition of which is intricate to say the least. Therefore, I shall depart. I am at the moment ill as I have an excruciating cough, and a swollen lymph node in my armpit at the time I am writing this. However, it will get better in a few days time. I shall take a couple days off as I am preparing for my graduation ceremony. Yes, I am writing these notes for fun and not from a syllabus or curriculum. I shall depart for now, and farewell.*

Date of Beginning - 10/06/24, 13:54.

Date of End - 16/06/24, 21:00.

3.2 June 21, 2024.

[09:22] The following subsection will involve congruence, and the ideal introduction to factoring and expanding terms.

3.2.1 Identical Polynomials

The idea of *congruence* means that they are equal or identical. In geometry, this is used more visually, like determining the congruence of two triangles only by their angle lengths. But for this topic, we do not involve geometry in our descriptive. We are going to focus more on about algebraic congruence; the congruence found in polynomials. The definition of *identical* or *identity* for our use is the same idea as congruence. However, congruence uses $\cong$ as its symbol rather than identity being $\equiv$.

Definition 3.4 (Polynomial Identity). $A \equiv B$ if they are reducible to each other; by using the six field axioms, axioms of equality, or whatever aforementioned to form the same equation.

Example 3.11. The following:

1. $x - 11 \equiv 3x + 7 - 2(x + 2)$
2. $xy + x \equiv x(y + 1)$
3. $y^2 - x^2 \equiv (y + x)(y - x)$

Two polynomials are *identically equal* if they are fully reduced/solved and are equal. $3x+1 \equiv 3x+1$ is identically equal, but $x(2 + 1) + 1 \equiv 3x + 1$ is not. We can deduce a theorem from corresponding coefficients:

Theorem 3.1 (Corresponding Coefficients Identity). If a polynomial in variable α or Greek letter alpha (to represent any single variable polynomial) are identically equal;

$$a_0\alpha^n + a_1\alpha^{n-1} + \cdots + a_n \equiv b_0\alpha^n + b_1\alpha^{n-1} + \cdots + b_n$$

$$\Rightarrow a_0 = b_0, a_1 = b_1, \dots, a_n = b_n$$

We can use this to deduct on solving problems like related to the following:

Example 3.12. $ax^2 - 3x + c \equiv x^2 - bx + 19$. Find a , b , and c .

We can deduce from **Theorem 3.12** that $a = 1$, $b = -3$, $c = 19$.

Another theorem is used if more than one variable exists with each having their own powers.

Theorem 3.2.

$$a + bx + cy + dxy + ex^2 + fx^2y + gx^2y^2 + hxy^2 + iy^2 + \dots$$

$$\equiv a' + b'x + c'y + d'xy + e'x^2 + f'x^2y + g'x^2y^2 + h'xy^2 + i'y^2 + \dots$$

$$\Rightarrow a = a', b = b', c = c', d = d', \dots$$

if they are identically equal polynomials.

Identical polynomials hold the same properties with the *Transitivity*, *Reflexive*, and *Symmetry* axioms respectively. Hence,

- $A \equiv B \Rightarrow B \equiv A$
- $A \equiv A$
- $A \equiv B \wedge B \equiv C \Rightarrow A \equiv C$

The idea that they hold with *Transitivity*, *Reflexive*, and *Symmetry* axioms, we can deduct that the *Addition Property* and *Multiplication Property* equality axioms must also hold. They are considered as *transformations* as they change the equation. However, since we consider identical kinds, we must assume that the other end must also be transformed the same as the first. So, we consider the next theorem:

Theorem 3.3 (Transformations). If a supposed identity $A \equiv B$ and a defined/known identity $C \equiv D$ exists and if the congruences can derive each other or one reverses to the former, then the supposed identity $A \equiv B$ holds.

A special corollary¹¹ can be thought:

Corollary 3.1. If a transformation occurs on $A \equiv B$, it must hold its congruence.

Using this with an example:

Example 3.13. Seen on the following:

$$a \equiv b \text{ is true.} \quad (17)$$

$$a + c \equiv b + c \text{ is true by the } \textit{Addition Property} \text{ equality axiom.} \quad (18)$$

$$(a + c) - d \equiv b + c \text{ is not true.} \quad (19)$$

(17) and (18) are both related by reversing a process, subtracting c will give the given. Hence, (17) and (18) hold true in that (18) is a transformation related to (17). However (19) does not have a process that gives back (18), hence it is not a transformation related to (18).

3.2.2 Basis of Factoring

Ideally, *factoring* involves taking the common factor of the terms in a polynoial and expressing it as a product of two terms. Factoring can be of coefficients only, and with variables that are easily reducible. We can denote this as the following definition for factoring coefficients:

Definition 3.5 (Common Factor for Coefficients). If a polynomial P exists, wherein

$$P = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \cdots + a_2 x^2 + a_1 x + a$$

then the common factor of P is a .

If the common factors are reducible by a then the polynomial is *factored*. However, I must denote some notation first before we proceed more intuitively for more ease and intuition.

Definition 3.6 (Division Notation). If $a|b$, then "a divides b" or $\frac{b}{a}$.

We can now proceed forward with the following definition:

Definition 3.7 (Defining Factoring with Coefficients). If $a|P$, a is the *greatest common factor* if

$$\begin{aligned} a|P &= \frac{a_n x^n}{a} + \frac{a_{n-1} x^{n-1}}{a} + \cdots + \frac{a_2 x^2}{a} + \frac{a_1 x}{a} + \frac{a}{a} \\ \therefore a|P &= a_{n-1} x^n + a_{n-2} x^{n-1} + \cdots + a x^2 + x + 1 \end{aligned}$$

¹¹A corollary is a type of definition that is from a theorem or something of importance, that adds more definitive reasons for an explanation, but is less of importance but still needs to be noticed.

Example 3.14. Factor $16x + 36y$.

In the coefficients, the common factors in 16 and 36 is 4. Reduce 16 and 36 to 4 and 9. Factor them out, we get $\boxed{4(4x + 9y)}$. We can prove this by doing the distributive property:

$$\begin{aligned} 4(4x + 9y) &= 4(4x) + 4(9y) \\ &= 16x + 36y \end{aligned}$$

Seen from the example, we take the greatest common factor of 16 and 36 and reduce them, and put them as a product of the greatest common factor and the factored polynomial. We can express this from our theorem, as we can get the same polynomial back. This process is called *expanding terms*.

Theorem 3.4 (Expanding Terms). If $a|P$ exists, wherein a is the GCF, $(a|P)(a) = P$.

Proof. Assume $a|P$ exists. If,

$$\begin{aligned} a|P &= \frac{a_n x^n}{a} + \frac{a_{n-1} x^{n-1}}{a} + \cdots + \frac{a_2 x^2}{a} + \frac{ax}{a} + \frac{a}{a} \\ &\Rightarrow a_{n-1} x^n + a_{n-2} x^{n-1} + \cdots + ax^2 + x + 1 \end{aligned}$$

by Definition 3.7. If we assume by *Distributivity*:

$$\begin{aligned} (a|P)(a) &= a \left(\frac{a_n x^n}{a} + \frac{a_{n-1} x^{n-1}}{a} + \cdots + \frac{a_2 x^2}{a} + \frac{ax}{a} + \frac{a}{a} \right) \\ &\Rightarrow a \left(\frac{a_n x^n}{a} \right) + a \left(\frac{a_{n-1} x^{n-1}}{a} \right) + \cdots + a \left(\frac{a_2 x^2}{a} \right) + a \left(\frac{ax}{a} \right) + a \left(\frac{a}{a} \right) \end{aligned}$$

If we assume in our lemma¹² for our definition of cancelling:

Lemma 3.5. $a \cdot \frac{b}{a} = b$.

$$\therefore a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + ax + a$$

□

Example 3.15. Show that $4(x^2 - 11) + 3 \equiv 4x^2 - 41$.

Using *Distributivity*,

$$\begin{aligned} 4(x^2 - 11) + 3 &\equiv 4x^2 - 41 \\ 4(x^2) - 4(11) + 3 &\equiv 4x^2 - 41 \\ 4x^2 - 44 + 3 &\equiv 4x^2 - 41 \\ 4x^2 - 41 &\equiv 4x^2 - 41 \end{aligned}$$

is true

Note that, the common factor of two powers is the smallest one possible that exists between terms.

¹²A *lemma* is a proposition, or statement that is used to prove a theorem, and is a generally accepted proposition. It is also known as an "auxiliary theorem."

Theorem 3.6. If there exists a power n and a smallest power m in P , then

$$P = ax^n + bx^{n-1} + cx^{n-2} + \dots + zx^m$$

x^m is the common factor. Since m is the smallest power, then $m < n$.

Proof. Let $x^m | P$.

$$\begin{aligned} x^m | P &= \frac{ax^n}{x^m} + \frac{bx^{n-1}}{x^m} + \frac{cx^{n-2}}{x^m} + \dots + \frac{zx^m}{x^m} \\ &\Rightarrow \left(\frac{a}{1} \cdot \frac{x^n}{x^m} \right) + \left(\frac{b}{1} \cdot \frac{x^{n-1}}{x^m} \right) + \left(\frac{c}{1} \cdot \frac{x^{n-2}}{x^m} \right) + \dots + \left(\frac{z}{1} \cdot \frac{x^m}{x^m} \right) \\ &\therefore ax^{n-m} + bx^{(n-1)-m} + cx^{(n-2)-m} + \dots + z \end{aligned}$$

We remember that, $m < n \therefore n - m > 0$ and will not yield any non-integral expressions. $\square$

Example 3.16. Factor $4x^5 - 11x^3 + x^2$.

The variables x^5 , x^3 , and x^2 have the common factor of the smallest power, x^2 . Reduce x^5 , x^3 and x^2 . By using **Theorem 1.20**, then $\frac{x^5}{x^2} = x^{5-2} = x^3$, etc. $x^2(4x^3 + 11x + 1)$ is our factored form of the former. We can prove it by *Distributivity*, and it is left as a reader's exercise.

Example 3.17. Factor $20x^4 - 8x^3 - 16x^2 + 12x$.

The variables x^4 , x^3 , x^2 , x have the common factor of the smallest power x . Reduce them by **Theorem 1.20**. The coefficients 20, 8, 16, 12 have the common factor of 4. Divide them all by 4, we get 5, 2, 4, and 3. Multiply both factors of the variable and the coefficients, making our entire factor $4x$. Our factored polynomial would be $4x(5x^3 - 2x^2 - 4x + 3)$.

Example 3.18. Factor $14x^2y - 10xy$.

The variables x^2 , x and y have the common factors x and y respectively, together as xy . The coefficients 14 and 10 have the common factors for 2. Together as $2xy$, divide all terms by $2xy$. Our factored polynomial would be $2xy(7x - 5)$.

3.2.3 Common Results in Factoring

Various results are obtained by factoring. However, some results are more likely to appear often and are needless to say memorized to figure patterns between equations.

Definition 3.8 (Difference of Two Squares). If x and y are $x^2 - y^2$, then $x^2 - y^2 = (x + y)(x - y)$.

This result is most prominent in factoring and is a basis on how it works. However, we must assume that $x, y \in \mathbb{Q}$ for now. Most commonly we do not associate this factorization to be bounded in all real numbers, however this also works for if it was bounded to such.

IF BOUNDED BY $\mathbb{Q}$: The following are a list of examples:

Example 3.19. The following are examples:

1. Factor $x^2 - 9$.

$$\begin{aligned} x^2 - 9 &= x^2 - (3)^2 \\ &\Rightarrow (x + 3)(x - 3) \end{aligned}$$

2. Factor $a^2b^2 - 100$.

$$\begin{aligned} a^2b^2 - 100 &= a^2b^2 - (10)^2 \\ &\Rightarrow (ab + 10)(ab - 10) \end{aligned}$$

3. Factor $27x^3 - 12x$.

It may appear that we we can not factor it, but we can by *deducing a pattern*. If we realize that the common factor of this equation is $3x$, then we get $3x(9x^2 - 4)$. We can now factor it in the inner groupings:

$$\begin{aligned} 27x^3 - 12x &\equiv 3x(9x^2 - 4) \\ \therefore 3x(9x^2 - 4) &= 3x((3)^2x^2 - (2)^2) \\ &\Rightarrow 3x(3x + 2)(3x - 2) \end{aligned}$$

4. Factor $\frac{1}{16}x^2 - 1$.

$$\begin{aligned} \frac{1}{16} &= \left(\frac{1}{4}\right)^2 \therefore \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{4 \cdot 4} \\ \frac{1}{16}x^2 - 1 &= \left(\frac{1}{4}\right)^2 - (1)^2 \\ &\Rightarrow \left(\frac{1}{4}x + 1\right) \left(\frac{1}{4}x - 1\right) \end{aligned}$$

However, if we extend the domain to all real numbers, we can expect irrational roots to be in the mix. But for the sake of our explanation, we will save it for when we will get to properties of radicals.

The following is also commonly expected as a result if you were to square a binomial.

Definition 3.9 (Perfect-Square Trinomial). If x and y are $(x \pm y)^2$, then $(x \pm y)^2 = x^2 \pm 2xy + y^2$.

Remark 1. We should note by the description of $\pm$ is:

$$\begin{aligned} (x + y)^2 &\Rightarrow x^2 + 2xy + y^2 \\ (x - y)^2 &\Rightarrow x^2 - 2xy + y^2 \end{aligned}$$

The converse is also true of Definition 3.9:

Theorem 3.7. (Converse of Definition 3.9) $x^2 \pm 2xy + y^2 \Rightarrow (x \pm y)^2$

Proof. We will apply this for if $\pm$ is $+$. Let $x^2 + 2xy + y^2$ exist.

$$\begin{aligned} x^2 + 2xy + y^2 &= (x \cdot x) + (xy + xy) + (y \cdot y) \\ (x \cdot x + xy) + xy + (y \cdot y) &= x(x + y) + xy + (y \cdot y) \\ x(x + y) + (xy + (y \cdot y)) &= x(x + y) + y(x + y) \\ &\Rightarrow (x + y)(x + y) \\ &= \boxed{(x + y)^2} \end{aligned}$$

□

We can provide some examples:

Example 3.20. Factor $x^2 + 6x + 9$.

$$\begin{aligned} x^2 + 6x + 9 &= x^2 + 2(1)(3)(x) + (3)^2 \\ &\Rightarrow (x + 3)^2 \end{aligned}$$

Example 3.21. Factor $9x^2 + 42xy + 49y^2$.

$$\begin{aligned} 9x^2 + 42xy + 49y^2 &= (3)^2 + 2(3)(7)(x)(y) + (7)^2y^2 \\ &\Rightarrow (3x + 7y)^2 \end{aligned}$$

Example 3.22. Factor $(12q^2 - 60q + 75)$.

It may seem that there is no proper perfect-square trinomial that can fit this trinomial. However, the GCF of 12, 60, and 75 is 3 itself. We can begin by using general factoring first, then factoring using our method.

$$\begin{aligned} 12q^2 - 60q + 75 &= 3(4q^2 - 20q + 25) \\ 3((2)^2 - 2(2)(5)(q) + (5)^2) &\Rightarrow 3(2x - 5)^2 \end{aligned}$$

All of these factorizations can be expanded through or Distributivity. **Example 3.22** can be applied to all of the common results seen here, like the difference of two squares:

Example 3.23. Factor $32k^2 - 8$.

The GCF of 32 and 8 is 8 itself. Factor out:

$$\begin{aligned} 32k^2 - 8 &= 8(4k^2 - 1) \\ 8((2)^2k^2 - (1)^2) &\Rightarrow 8(2k + 1)(2k - 1) \end{aligned}$$

The following result is, in turn, another common result in factoring. It is as similar to the Difference of Squares, but it involves powers of three, or the *cube*. The following:

Definition 3.10 (Difference of Two Cubes). If x and y are $x^3 - y^3$, then $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$.

Definition 3.11 (Sum of Two Cubes). If x and y are $x^3 + y^3$, then $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$.

They can be verified by Distributivity. The topic on denoting on sums and differences of two powers is up for a future topic to handle.

Example 3.24. Factor $27a^3 - 8a^3$.

$$\begin{aligned} 27b^3 - 8a^3 &= (3)^3a^3 - (2)^3a^3 \\ (3a - 2b)((3a)^2 + (3a)(2b) + (2b)^2) &\Rightarrow (3a - 2b)(9a^2 + 6ab + 4b^2) \end{aligned}$$

Example 3.25. Factor $x^3 + 64$.

$$\begin{aligned} x^3 + 64 &= x^3 + (4)^3 \\ (x + 4)(x^2 - 4x + (4)^2) &\Rightarrow (x + 4)(x^2 - 4x + 16) \end{aligned}$$

Example 3.26. Factor $24x^3 + 81y^3$.

Notice the GCF of 24 and 81 is 3. We can begin to factor by first performing general factoring:

$$\begin{aligned} 24x^3 + 81y^3 &= 3(8x^3 + 27y^3) \\ 3((2)^3x^3 + (3)^3y^3) &= 3(2x + 3y)((2x)^2 - (2x)(3y) + (3y)^2) \\ &\Rightarrow 3(2x + 3y)(4x^2 - 6xy + 9y^2) \end{aligned}$$

FINISHING REMARKS – *This entire topic has been on the scope on how identical polynomials work, and the general basis of factoring as a whole. This entire topic has been on delay for weeks because of some demotivation but it is still completed. I also went through a graduation ceremony, and some happy moments with my friends which has taken over my mind rather than typesetting this document itself (which is irresponsible of me so I shall apologize, but I can't have a promise it won't be consistent.) The next topic will be about more on factoring, and the usage of the "FOIL" method. However, factoring is not done within just this basis, however, it is more thoroughly explained through more methods but it is not conventional to bring this approach to now because we do not have the requisite knowledge to perform such task. However, the grounds that this part has given has been excruciatingly important to our topic in algebra. We can now proceed to the next couple of days for the new topics ahead of us. I will now depart for now, and farewell.*

Date of Beginning - 21/06/24, 09:22.

Date of End - 11/07/24, 22:19.

This document never came to be because the way I wrote this document was not efficient, and I was fourteen when I wrote this. It will remain in this premature state.